DSCH-Net: Diffusion-State-Contextual Hybrid Network for Physics-Inspired and Direction-Aware Dehazing of Remote Sensing Imagery

Naveed Sultan, Mansoor Hayat, *Member, IEEE*, and Santitham Prom-on, *Member, IEEE*

***Abstract*—Atmospheric haze reduces contrast and suppresses fine structures in remote sensing imagery, degrading the reliability of downstream mapping and detection tasks. Existing convolutional neural network (CNN) and attention-based dehazing methods often struggle to capture long-range dependencies efficiently and may introduce color shifts or halo artifacts under nonuniform haze. Diffusion-state contextual hybrid network (DSCH-Net) is proposed as a physics-inspired and direction-aware state-space network for single-image dehazing. The framework combines a four-directional state-space operator for linear-time global context modeling with a partial differential equation (PDE)-driven diffusion module that injects explicit gradient–divergence priors to refine structural details. A residual multidilation unit recovers fine textures, while a selective fusion gate stabilizes encoder–decoder interactions to preserve spatial consistency. Extensive experiments on synthetic and real remote-sensing benchmarks show that DSCH-Net achieves consistent state-of-the-art performance, reaching up to 32.20-dB peak signal-to-noise ratio (PSNR) and 0.980 structural similarity index (SSIM) on RSID and delivering strong results across RICE1/2, Haze1K, and DHID. Visual evaluations further demonstrate clearer sky gradients, sharper boundaries, and improved preservation of thin structures compared to recent CNN and Transformer models. DSCH-Net offers a compact, end-to-end, and resolution-scalable solution suitable for practical remote-sensing workflows under adverse weather conditions. Code and data are available at https://github.com/CPEKMUTT/DSCH-Net**

***Index Terms*—Diffusion prior, global context, physics-guided restoration, single-image dehazing, state-space modeling.**

I. Introduction

IMAGE dehazing is essential for reliable interpretation of outdoor and remote sensing imagery, where haze reduces contrast, obscures fine structures, and degrades the performance of downstream tasks such as classification, detection, tracking, and semantic segmentation [1]. The challenge is more pronounced in nonhomogeneous haze, where spatially varying density and similarity to natural sky tones make it difficult to separate atmospheric effects from true scene content [2], [3]. As a result, recovering true colors, restoring thin structures, and stabilizing global context remain difficult.

Early prior-based approaches made initial progress but proved unreliable in real-world conditions. These methods often relied on assumptions of uniform haze and simple scattering behavior and therefore struggled when haze varied across space [4], [5]. Subsequent advances in deep learning and convolutional neural network (CNN)-based networks improved robustness and modeling capacity [6]. However, two persistent gaps remain. First, fusing global context with local detail is difficult, and high perceptual quality is not guaranteed for remote sensing imagery [7]. Second, pixel-oriented training losses tend to over-smooth textures, reducing the visibility of delicate structures [8], [9], [10], [11], [12]. Generative adversarial network (GAN)-based designs partially improve perceptual sharpness [13], [14], [15], [16], but may distort geometry near edges. Although controllable architectures mitigate this tension [17], the balance between structure preservation and artifact suppression remains delicate.

Transformers offer improved global reasoning [18], [19], yet their computational cost limits scalability to large remote-sensing images. Multiscale fusion remains complex in practice, and advanced attention mechanisms, while helpful, do not fully eliminate residual artifacts or color inconsistencies [20], [21]. Nonhomogeneous haze continues to induce halos, sky–object color shifts, and incomplete structural recovery.

For example, in aerial scenes with bright sky or water adjacent to high-contrast urban structures (e.g., coastlines, building roofs, and road boundaries), many dehazing models over-estimate transmission near the skyline, leading to halo rings and bluish/greenish color bias, while thin structures such as narrow roads, power lines, and small rooftop edges remain partially suppressed or over-smoothed [4], [5], [20], [51].

As illustrated in Fig. 1, atmospheric particles attenuate direct radiance from the ground and inject airlight into the sensor, which reduces contrast and obscures structural details in remote-sensing imagery. Motivated by the limitations of conventional priors, heavy attention modules, and high-sampling-cost diffusion pipelines, this article proposes a physics-guided state-space formulation that avoids quadratic attention while remaining compact and scalable. The objec-

Received 30 December 2025; revised 17 February 2026 and 14 March 2026; accepted 14 April 2026. Date of publication 20 April 2026; date of current version 29 April 2026. This work was supported in part by the King Mongkut's University of Technology Thonburi (KMUTT) Staff Research Grant for Fiscal Year 2025–2026. *(Corresponding author: Santitham Prom-on.)*

Naveed Sultan and Santitham Prom-on are with the Department of Computer Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi, Bangkok 10140, Thailand (e-mail: naveed.sult@kmutt.ac.th; santitham.pro@kmutt.ac.th).

Mansoor Hayat is with the Section of Neurosurgery, Department of Surgery, Rady Faculty of Health Sciences, University of Manitoba, Winnipeg, MB R3T 2N2, Canada (e-mail: mansoor.hayat@umanitoba.ca).

Digital Object Identifier 10.1109/TGRS.2026.3685508

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

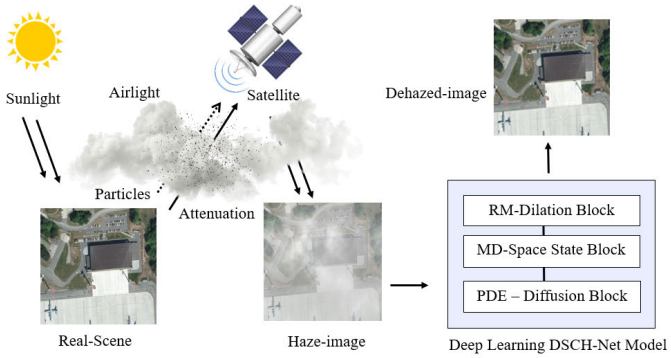


Fig. 1. Overview of the remote sensing dehazing process. Atmospheric haze introduces attenuation and airlight, degrading the observed scene and suppressing structural details. DSCH-Net restores the underlying clean image by modeling physics-guided diffusion, multidirectional state-space interactions, and multiscale texture enhancement.

tive is to stabilize color, capture long-range dependencies efficiently, and recover crisp textures. Instead of relying on handcrafted priors, the proposed design embeds physical cues implicitly through learnable operators that generalize across diverse scenes and haze conditions.

To address these challenges, a physics-guided encoder–decoder architecture equipped with selective skip fusion is developed. A multidirectional state-space block (MD-SSB) scans features horizontally and vertically in both directions, enabling long-range interaction with linear computational cost. A partial differential equation (PDE)-diffusion block introduces a learnable diffusion update based on discrete spatial differences and a gated step, suppressing color casts and noise while preserving edges under nonuniform haze. A residual multidilation block (RMDB) strengthens texture recovery using parallel dilated paths, and a selective fusion gate stabilizes encoder–decoder merging to reduce halos at bright sky–object boundaries.

The main contributions of this work are summarized as follows.

1) Diffusion-state contextual hybrid network (DSCH-Net) is introduced as a physics-guided encoder–decoder architecture that unifies multidirectional state-space scanning with a learnable diffusion process for stable and artifact-resistant dehazing.
2) A dedicated physics-guided diffusion block (PDE-DB) is developed that performs gated, data-driven diffusion using discrete spatial differences, improving color fidelity and edge preservation under nonhomogeneous haze.
3) A RMDB is designed to recover fine textures and thin structures after global context modeling, and a selective fusion gate is employed to stabilize encoder–decoder feature exchange, preserving spatial consistency and reducing halo artifacts.
4) Performance improvements are demonstrated over recent CNN, Transformer, and state-space baselines on synthetic and real datasets, with strong visual recovery of sky gradients, thin structures, and small text.

II. Related Work

Remote-sensing dehazing has been explored through physical priors, convolutional architectures, Transformer and hybrid-attention models, frequency and dual-domain fusion, state-space networks, and diffusion/generative approaches. Each line of work advances the field but retains limitations in large-scale, nonhomogeneous scenes.

A. Physical Priors and Atmospheric Models

Classical methods estimate transmission and airlight by inverting simplified atmospheric models. The dark channel prior and the color attenuation prior [4], [5] remain effective under mild and near-uniform haze but tend to fail in bright sky regions and spatially varying haze thickness, often producing color shifts and halos around edges. Remote-sensing variants that incorporate guided transmission maps, handcrafted constraints, or scene-specific heuristics [22], [23] improve robustness in certain acquisition conditions, but remain sensitive to sensor calibration and scene content and cannot generalize across the diverse illumination and land-cover distributions encountered in practice.

B. Convolutional Architectures

CNN-based approaches learn haze removal directly from paired data and strengthen spatial detail through multiscale encoders, high-resolution branches, and skip connections. Early U-Net variants and high-resolution refinements [24], [25], [26], [27]. enhanced edge and texture recovery for aerial and satellite imagery. However, convolutional operations are inherently local: even deep stacks with large receptive fields struggle to propagate global illumination cues and maintain color consistency across kilometer-scale orthomosaics. This locality leaves residual halos that degrade color fidelity over large homogeneous regions and result in incomplete structural recovery under nonhomogeneous haze. The need for efficient global context modeling without abandoning the spatial precision that CNNs provide is one of the central motivations for incorporating directional state-space scanning in DSCH-Net.

C. Transformers, Hybrid Attention, and Frequency Models

Transformer-based image restoration addressed the global context limitation of CNNs by enabling long-range dependency modeling through self-attention. General-purpose models such as IPT, SwinIR, Restormer, and Uformer [28], [29], [30], [31] improved global context aggregation and chromatic stability. Remote-sensing dehazing variants with cross-striping, proxy tokens, and cascaded global–local attention [20], [32] reduced computational cost while preserving contextual cues. Hybrid designs combining bidirectional CNN–Transformer modulation [33], [34] further enhanced texture quality. However, standard self-attention incurs quadratic complexity in the number of spatial tokens, making it prohibitive for high-resolution RS inference without aggressive window partitioning or token reduction. Multistage Transformer pipelines with dense skip connections are also computationally heavy, limiting practicality in operational RS

workflows. Critically, even with global attention, no existing Transformer formulation explicitly models the physical process of haze formation. Another complementary direction explicitly fuses spatial and frequency cues. Dual-branch pipelines that combine detail and chromaticity, or spatial and frequency information [35], [36], improve high-frequency reconstruction while stabilizing color. Wavelet- and frequency-enhanced networks, dynamic channel reweighting, and robust edge operators [37], [38] raise PSNR/SSIM and perceptual quality on RS benchmarks.

D. State-Space Models

State-space models (SSMs) have recently emerged as a compelling path to global dependency modeling without the quadratic cost of attention. By parameterizing linear recurrent systems over spatial sequences, SSMs achieve linear-time axial scans that propagate information across the full image extent. Vision-Mamba units and U-shaped Mamba variants [39] achieve competitive RS dehazing by efficiently mixing local and global cues through directional recurrences. Physics-decoupled or scale-variant SSM formulations [40] further stabilize large-scale color by separating physical haze components from learned representations. Nonetheless, existing SSM-based models typically scan in one or two fixed directions, which can leave anisotropic haze gradients under-corrected. More importantly, SSM recurrences provide efficient global mixing but do not inject explicit physics-based priors for haze suppression, which can be unreliable under severe spatial variation. These models also require careful skip fusion design to prevent halos and maintain texture fidelity in scenes with strong haze gradients, an aspect that is not fully addressed in current formulations. DSCH-Net builds on the efficiency of SSMs while addressing these gaps through four-directional scanning, a dedicated PDE-DB, and a selective fusion gate.

E. Diffusion and Generative Approaches

Generative models approach haze removal from the perspective of perceptual quality and distributional realism. GAN-based dehazers [13], [41] sharpen edges and synthesize plausible textures but frequently distort geometry properties. Diffusion-based dehazing [42] introduces Fourier-aware priors and coarse-to-fine sampling and improves stability. Residual and dual-stage diffusion designs [43], [44], [45] reduce sampling steps, while lightweight and encoder-free generators [46] reduce parameter count. Yet, many diffusion pipelines require long sampling chains in operational workflows. More fundamentally, these models often plateau under severe nonhomogeneous haze, leaving thin structures under-restored, because the diffusion process does not explicitly model the spatial variation of haze density. Inspired by diffusion priors, DSCH-Net incorporates a deterministic, learnable PDE-DB that injects edge-aware smoothing behavior directly into the feature update, capturing the physical insight of diffusion without the sampling overhead.

F. Datasets and Evaluation Protocols

Remote-sensing dehazing benchmarks have progressively targeted more realistic and challenging conditions. SateHaze1K [47] provides haze-level annotations for synthetic RS pairs across thin, moderate, and thick densities. Real-world collections such as RRSHID [48] introduce sensor-calibrated hazy–clean pairs, while DHID [49] focuses on challenging in-the-wild haze. Broader real-world dehazing suites [50] test generalization beyond synthetic compositions. Across all these datasets, persistent failure modes are consistently reported: color casts, halos near skylines, loss of small structures, and tiling seams during high resolution.

G. Recent RS-Focused Designs

Recent models combine attention, bilateral learning, and ultrahigh-resolution encoders to reduce artifacts and improve reconstruction [7], [27], [51], [52]. Guided transmission Transformers, hybrid high-resolution networks, and cascaded attention refine both global and local priors. Modulation across CNN/Transformer tokens, dual-domain interaction, and cascaded global–local attention [33], [34], [36] mitigate oversmoothing and color drift. Lightweight or frequency-guided diffusion models [42], [43], [46] accelerate inference but often struggle in dense, nonhomogeneous haze or around sky–object edges.

Across these directions, four interrelated challenges remain: 1) the need for efficient long-range coupling that scales linearly with image size; 2) physics-guided control over low-frequency haze and color diffusion robust to nonuniform thickness; 3) texture restoration that faithfully recovers wires, small text, and thin structures without halos; and 4) selective skip fusion that avoids ringing artifacts while remaining stable across sensors and seasons. The proposed DSCH-Net design targets these issues by combining multidirectional state-space scanning for linear-time global context, a learnable PDE-guided diffusion update for haze veil suppression, residual multidilation for texture recovery, and a selective fusion gate for artifact-free skip merging. As demonstrated in the experiments, this combination closes much of the observed performance gap on synthetic and real RS benchmarks while preserving practical efficiency [53], [54], [55], [56], [57].

III. Method

DSCH-Net is designed to jointly address three key challenges in remote-sensing dehazing: 1) capturing long-range context over large scenes; 2) stabilizing the low-frequency haze veil and color; and 3) recovering fine textures and thin structures without halos. To this end, the network combines a PDE-DB motivated by the need to suppress spatially varying airlight through learnable, edge-aware smoothing rather than handcrafted priors, a multidirectional state-space module that provides linear-time global context by scanning features along four spatial axes, and an RMDB that recovers fine details attenuated after global haze suppression. These three components are embedded inside an encoder–decoder backbone with selective skip fusion, which prevents halo formation at

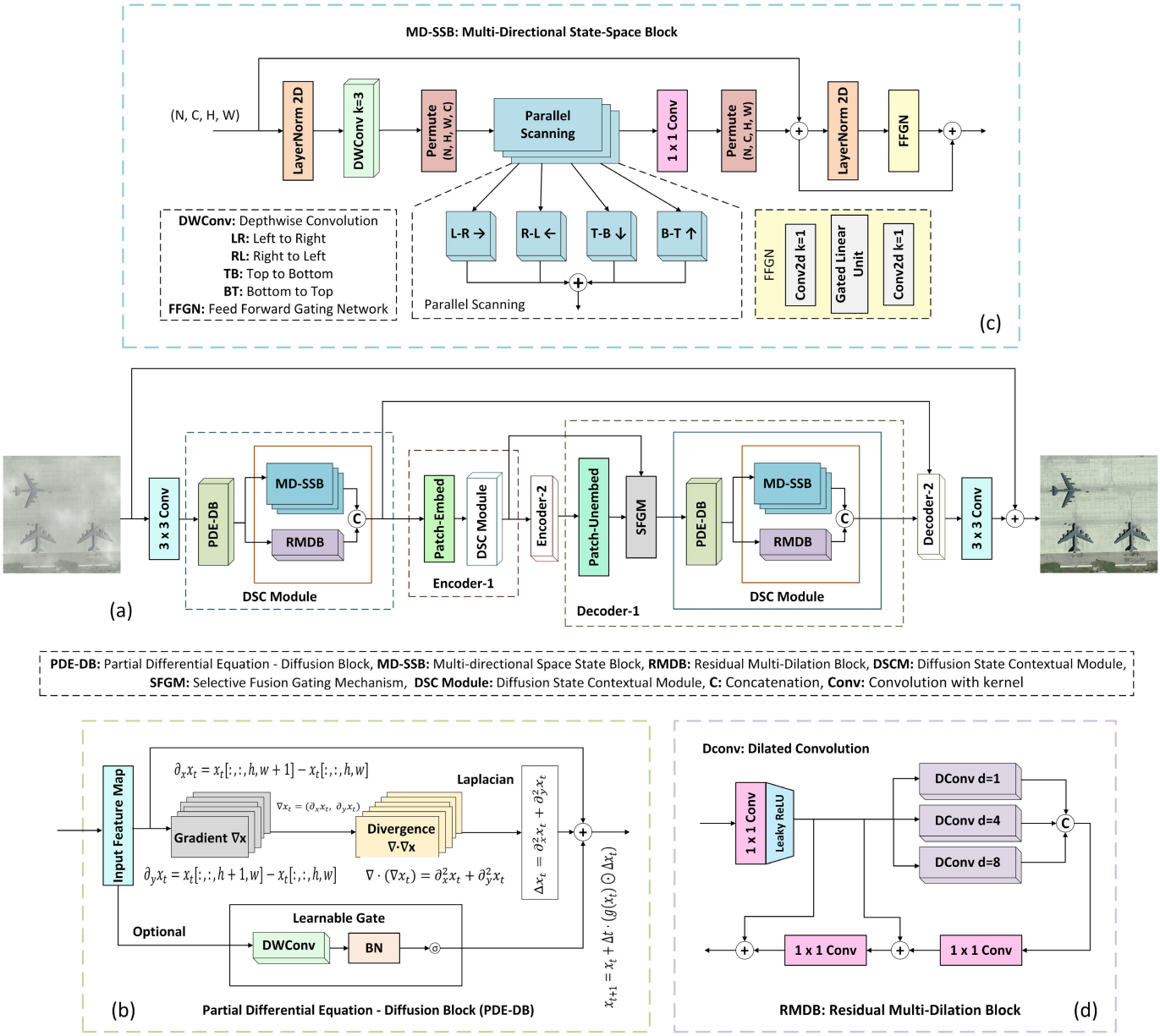


Fig. 2. Overview of DSCH-Net. (a) Full encoder–decoder framework, which applies the proposed diffusion–state–contextual (DSC) module at multiple scales. Each stage integrates physics-guided diffusion, multidirectional state-space scanning, and multidilation texture enhancement, while the selective fusion gating mechanism (SFGM) stabilizes skip connections. (b) PDE-DB introduces a learnable diffusion step based on discrete gradient, divergence, and Laplacian operators, enabling haze-veil suppression and color stabilization while preserving edges. (c) MD-SSB performs four parallel axial scans (left–right, right–left, top–bottom, and bottom–top) using linear state-space recurrences, providing efficient long-range dependency modeling. (d) RMDB aggregates multiscale features through parallel dilated convolutions to recover thin structures, textures, and small details.

bright sky–object boundaries by adaptively gating encoder and decoder feature streams as illustrated in Fig. 2.

Given an input image $\mathbf{I} \in \mathbb{R}^{B\times 3\times H\times W}$, where B is the batch size, H and W denote spatial height and width, a 3×3 convolution projects $\mathbf{I}$ into an initial feature representation $\mathbf{X}_0 \in \mathbb{R}^{B\times C\times H\times W}$ with C channels. At each scale, the features are refined by a DSC, which sequentially applies the PDE-DB, MD-SSB, and RMDB

$$\mathbf{Z} = \mathrm{DSC}(\mathbf{X}) = \mathrm{RMDB}(\mathrm{MD}-\mathrm{SSB}(\mathrm{PDE}-\mathrm{DB}(\mathbf{X}))) \quad (1)$$

where $X \in \mathbb{R}^{B\times C\times H\times W}$ denotes the input feature tensor, PDE-DB(·) represents the PDE-DB, MD-SSB(·) denotes MD-SSB, and RMDB(·) is RMDB. DSC(·) denotes the sequential composition of these three operators.

A. Physics-Guided Diffusion Block

The PDE-DB aims to remove the low-frequency haze veil and stabilize color while preserving edges. This design explicitly targets spatially nonuniform haze, in which local regions may require different diffusion strengths to suppress haze while maintaining boundaries and thin structures. Instead of applying diffusion directly to the RGB image, the operation is performed on intermediate features, allowing the network to learn where and how strongly diffusion should be applied.

Let $X_{i,j} \in \mathbb{R}^C$ denote the feature vector at spatial location (i, j), where $i \in \{1, \ldots, H\}$ and $j \in \{1, \ldots, W\}$. Forward finite differences are computed with reflective padding at the image boundaries

$$\partial_x \mathbf{X}_{i,j} = \mathbf{X}_{i,j+1} - \mathbf{X}_{i,j} \tag{2}$$

$$\partial_y \mathbf{X}_{i,j} = \mathbf{X}_{i+1,j} - \mathbf{X}_{i,j}. \tag{3}$$

The spatial gradient is defined as $\nabla X = [\partial_x X, \partial_y X] \in \mathbb{R}^{B\times C\times H\times W\times 2}$. The discrete Laplacian is computed as the divergence of the gradient, $\Delta X = \nabla \cdot \nabla X$, using backward differences for numerical stability

$$\mathbf{X}^+ = \mathbf{X} + \alpha \odot \Delta \mathbf{X} + \nabla \cdot (\kappa(\mathbf{X}) \odot \nabla \mathbf{X}) \tag{4}$$

where $X^+ \in \mathbb{R}^{B\times C\times H\times W}$ denotes the diffused feature map, $\alpha \in \mathbb{R}^C$ is a learnable per-channel diffusion step size, $\odot$ denotes elementwise multiplication with broadcasting, and $\kappa(X) \in \mathbb{R}^{B\times C\times H\times W}$ is a spatially adaptive conductance map that modulates diffusion strength.

Algorithm 1 PDE-DB

Input: Feature map $X \in \mathbb{R}^{B\times C\times H\times W}$, step $\Delta t > 0$, flags `use_gate`, `use_anisotropic`
Output: Updated feature map X^+

Finite differences: $\partial_x X_{i,j} \leftarrow X_{i,j+1} - X_{i,j}$, $\partial_y X_{i,j} \leftarrow X_{i+1,j} - X_{i,j}$ ▷reflect padding
$\nabla X \leftarrow [\partial_x X, \partial_y X]$
$(\nabla \cdot V)_{i,j} \leftarrow V_{xi,j} - V_{xi,j-1} + V_{yi,j} - V_{yi-1,j}$
$\mathcal{L}X \leftarrow \nabla \cdot \nabla X$ ▷discrete Laplacian
if `use_anisotropic` **then**
$\kappa \leftarrow \sigma(\mathrm{DWConv}_{3\times 3}(X))$
$U_{\text{aniso}} \leftarrow \nabla \cdot (\kappa \odot \nabla X)$
else
$U_{\text{aniso}} \leftarrow 0$
end if
if `use_gate` **then**
$G \leftarrow \sigma(\mathrm{DWConv}_{3\times 3}(X))$
$U \leftarrow G \odot \mathcal{L}X + U_{\text{aniso}}$
else
$U \leftarrow \mathcal{L}X + U_{\text{aniso}}$
end if
$X^+ \leftarrow X + \Delta t \cdot U$
return X^+

Although the diffusion step size is learned, the proposed formulation remains stable due to architectural constraints. Diffusion is applied to intermediate feature representations rather than raw intensities, and the update is embedded in a residual gated form with $\gamma \in (0, 1)$ which explicitly bounds the contribution of the diffusion term. In addition, the spatially adaptive conductance $\kappa(\mathbf{X})$ suppresses diffusion near strong gradients, preventing excessive updates at edges. Together, these mechanisms provide stable behavior in practice without requiring an explicit CFL-type constraint

$$\mathrm{PDE-DB}(\mathbf{X}) = \mathbf{X} + \gamma \odot \mathbf{X}^+, \quad \gamma \in (0, 1) \tag{5}$$

where $\gamma \in \mathbb{R}^C$ is a learnable per-channel residual scaling parameter that bounds the contribution of the diffusion update and stabilizes training. This design follows classical anisotropic diffusion [58], [59], which performs edge-preserving smoothing by promoting diffusion in homogeneous regions while suppressing it near strong gradients, but in this work, all coefficients are learned jointly with the network, allowing the model to adapt the diffusion behavior to different scenes and haze levels. The whole procedure, including optional isotropic/anisotropic variants, is summarized in Algorithm 1.

B. Multidirectional State-Space Block

While PDE-DB stabilizes low-frequency content locally, it also requires propagating information over large spatial extents to maintain consistent sky color and large-object boundaries. MD-SSB fulfills this role by scanning features in four directions with linear-time state-space recurrences.

For a 1-D sequence extracted along a row or column, let $\mathbf{u}_t \in \mathbb{R}^C$ be the input at step t. MD-SSB applies a linear SSM

$$\mathbf{s}_{t+1} = A\mathbf{s}_t + B\mathbf{u}_t, \quad \mathbf{y}_t = C\mathbf{s}_t + D\mathbf{u}_t \tag{6}$$

where $u_t \in \mathbb{R}^C$ is the input feature at spatial position t along a scan line, $s_t \in \mathbb{R}^C$ is the hidden state, and $y_t \in \mathbb{R}^C$ is the output. Matrices $A, B, C, D \in \mathbb{R}^{C\times C}$ are diagonal, enabling independent channelwise state transitions. The index t denotes the ordered spatial position along a row or column.

Run the state-space scan along four directions: left–right (LR), right–left (RL), top–bottom (TB), and bottom–top (BT), producing directional feature maps $\mathbf{Y}^{\mathrm{LR}}, \mathbf{Y}^{\mathrm{RL}}, \mathbf{Y}^{\mathrm{TB}}, \mathbf{Y}^{\mathrm{BT}} \in \mathbb{R}^{B\times C\times H\times W}$. Let $d \in \{\mathrm{LR}, \mathrm{RL}, \mathrm{TB}, \mathrm{BT}\}$ index the scan direction, and denote the corresponding directional output feature map by $\mathbf{Y}^d \in \mathbb{R}^{B\times C\times H\times W}$.

To combine the directional responses, a lightweight gating unit predicts adaptive fusion weights

$$\mathbf{G} = \sigma\left(\mathrm{Conv}_{1\times 1}\left(\left[\mathbf{Y}^{\mathrm{LR}}, \mathbf{Y}^{\mathrm{RL}}, \mathbf{Y}^{\mathrm{TB}}, \mathbf{Y}^{\mathrm{BT}}\right]\right)\right) \tag{7}$$

$$\mathbf{Y} = \sum_{d\in\{\mathrm{LR},\mathrm{RL},\mathrm{TB},\mathrm{BT}\}} \mathbf{G}_d \odot \mathbf{Y}^d \tag{8}$$

where $[\cdot]$ denotes channelwise concatenation, $\sigma(\cdot)$ denotes the sigmoid activation function applied elementwise, and $\odot$ denotes elementwise multiplication with broadcasting. The tensor $\mathbf{G} \in (0, 1)^{B\times 4C\times H\times W}$ contains direction- and channel-specific fusion weights and is partitioned into four channel groups $\mathbf{G} = [\mathbf{G}_{\mathrm{LR}}, \mathbf{G}_{\mathrm{RL}}, \mathbf{G}_{\mathrm{TB}}, \mathbf{G}_{\mathrm{BT}}]$, where each $\mathbf{G}_d \in (0, 1)^{B\times C\times H\times W}$ corresponds to one scan direction and modulates the associated directional feature map.

The fused feature map $\mathbf{Y}$ is then projected by a 1×1 convolution and combined with the input through a residual connection

$$\mathrm{MD-SSB}(\mathbf{X}) = \mathbf{X} + \phi \odot \mathrm{Conv}_{1\times 1}(\mathbf{Y}) \tag{9}$$

where $\phi \in \mathbb{R}^C$ is a learnable per-channel residual scaling parameter that controls the contribution of the fused directional response. Each directional scan has complexity $O(CHW)$, and the four scans run in parallel, resulting in overall linear complexity in the image size and avoiding the quadratic cost of self-attention. Algorithm 2 details the implementation, including prenormalization and a lightweight feed-forward refinement.

Algorithm 2 MD-SSB

Input: $X \in \mathbb{R}^{B\times C\times H\times W}$, $\mathcal{D} = \{\text{LR}, \text{RL}, \text{TB}, \text{BT}\}$
Output: Enhanced feature map Y
$Y_0 \leftarrow \text{DWConv}_{3\times3}(\text{LN2D}(X))$
Convert Y_0 to channels-last $\tilde{Y} \in \mathbb{R}^{B\times H\times W\times C}$
$\mathcal{S} \leftarrow \emptyset$
for all $d \in \mathcal{D}$ **do**
  Reshape $\tilde{Y}$ into 1D sequences along rows/columns
  Apply channelwise SSM scan: $s_{t+1} = As_t + Bu_t$, $y_t = Cs_t + Du_t$
  Reshape back to $B \times H \times W \times C$ and insert into $\mathcal{S}$
end for
Fusion: $\hat{Y} \leftarrow \sum_{S\in\mathcal{S}} S$ ▷or concatenate then 1×1 conv
$Z \leftarrow \text{Conv}_{1\times1}(\text{NCHW}(\hat{Y}))$; $X_1 \leftarrow X + \text{Drop}(Z)$
$T \leftarrow \text{LN2D}(X_1)$
$T \leftarrow \text{GEGLU}(\text{Conv}_{1\times1}(T))$
$T \leftarrow \text{Conv}_{1\times1}(T)$
$Y \leftarrow X_1 + \text{Drop}(T)$
return Y

C. Residual Multidilation Block

After diffusion and global mixing, some fine-scale details remain attenuated. RMDB focuses on restoring those features by aggregating responses from multiple dilation rates in a residual form.

Algorithm 3 RMDB

Input: $X \in \mathbb{R}^{B\times C_{in}\times H\times W}$, $\mathcal{D} = \{1, 4, 8\}$
Output: Refined feature map Y
$R \leftarrow \text{LeakyReLU}(\text{Conv}_{1\times1}(X))$
for all $d \in \mathcal{D}$ **do**
  $F_d \leftarrow \text{Conv}_{3\times3}^{\text{dil}=d}(R)$
end for
$F \leftarrow \text{Concat}_C(\{F_d\}_{d\in\mathcal{D}})$
$Z \leftarrow \text{Conv}_{1\times1}(F)$; $Z \leftarrow Z + R$
$Z \leftarrow \text{Conv}_{1\times1}(Z)$; $Y \leftarrow Z + R$
return Y

Starting from input $\mathbf{X}$, first compress channels with a 1×1 convolution and apply a nonlinearity. Three parallel dilated 3×3 convolutions with dilation factors $d \in \{1, 4, 8\}$ then extract features at different effective receptive fields

$$\mathbf{T}_d = \text{Conv}_{3\times3}^{(d)}\left(\text{LReLU}\left(\text{Conv}_{1\times1}(\mathbf{X})\right)\right) \tag{10}$$

where LReLU denotes the Leaky ReLU activation, defined as $\text{LReLU}(x) = x$ if $x \geq 0$ and αx otherwise, with negative slope $\alpha = 0.2$ is used throughout all experiments. Their sum is fused and projected back by a 1×1 convolution

$$\text{RMDB}(\mathbf{X}) = \mathbf{X} + \text{Conv}_{1\times1}(\mathbf{T}_1 + \mathbf{T}_4 + \mathbf{T}_8). \tag{11}$$

Short-range dilation ($d = 1$) preserves local texture, whereas larger dilations capture thin structures such as wires or building edges without increasing depth. The exact steps appear in Algorithm 3. Placed after MD-SSB, RMDB refines high-frequency content on top of globally consistent features.

D. Selective Fusion Gating Mechanism

Naïve concatenation or addition of encoder and decoder features can create halos at bright boundaries, especially where haze thickness changes abruptly. SFGM alleviates this by adaptively weighting both streams in channel and space. Given encoder feature $\mathbf{E}$ and decoder feature $\mathbf{D}$ at the same scale, first form a channel descriptor

$$\mathbf{q} = \text{GAP}([\mathbf{E}, \mathbf{D}]) \tag{12}$$

where GAP$(\cdot)$ denotes global average pooling over spatial dimensions, yielding $q \in \mathbb{R}^{B\times2C\times1\times1}$, and map it through a small MLP followed by a softmax to obtain two scalar gates per channel:

$$\boldsymbol{\pi} = \text{Softmax}(\text{MLP}(\mathbf{q})) \in \mathbb{R}^{B\times2\times1\times1} \tag{13}$$

where the softmax is applied along the encoder–decoder dimension, ensuring $\pi_1 + \pi_2 = 1$ for each channel. The channelwise fusion is

$$\text{SFGM}(\mathbf{E}, \mathbf{D}) = \boldsymbol{\pi}_1 \odot \mathbf{E} + \boldsymbol{\pi}_2 \odot \mathbf{D}. \tag{14}$$

Optionally, a shallow depthwise convolution on $[\mathbf{E}, \mathbf{D}]$ predicts a spatial mask that further suppresses inconsistent activations along edges. SFGM is inserted at every skip connection in Fig. 2, ensuring that encoder detail is reused only where it is consistent with the decoder's current estimate.

E. Encoder–Decoder Pipeline and Output

The encoder progressively reduces spatial resolution and increases channel capacity using patch embedding. For a feature map $\mathbf{X}_s$ at scale s, the next scale is obtained via

$$\mathbf{X}_{s+1} = \text{PE}(\text{DSC}(\mathbf{X}_s)) \tag{15}$$

where PE$(\cdot)$ denotes patch embedding implemented via nonoverlapping 2×2 spatial packing followed by a 1×1 projection. The decoder mirrors this process with patch unembedding and SFGM-based skip fusion. At scale s, let $\mathbf{E}_s$ denote the encoder feature and $\mathbf{Y}_{s+1}$ the deeper decoder feature, which is first upsampled via patch unembedding before fusion

$$\mathbf{Y}_s = \text{DSC}\left(\text{PU}\left(\mathbf{Y}_{s+1}\right) + \text{SFGM}\left(\mathbf{E}_s, \text{PU}\left(\mathbf{Y}_{s+1}\right)\right)\right) \tag{16}$$

where PU$(\cdot)$ denotes patch unembedding via nearest-neighbor upsampling and channel projection. At the final scale, a 3×3 convolution predicts a residual map $\widehat{\mathbf{R}}$, and the dehazed image is obtained by

$$\widehat{\mathbf{I}} = \mathbf{I} + \widehat{\mathbf{R}}. \tag{17}$$

F. Loss Function

To train DSCH-Net, a composite loss function is employed that combines pixelwise fidelity and perceptual consistency. The overall objective is designed to ensure accurate haze removal while preserving structural details and visual realism

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N}\left\|I_{\text{DH}}^{(i)} - I_{\text{GT}}^{(i)}\right\|_1 \tag{18}$$

TABLE I
COMPARISON OF DSCH-NET WITH STATE-OF-THE-ART MODELS ON THE STATEHAZE1K DATASET. ↑ INDICATES HIGHER IS BETTER. ↓ INDICATES LOWER IS BETTER

Method	Venue/Year	StateHaze1K-Thin			StateHaze1K-Moderate			StateHaze1K-Thick			StateHaze1K-Average		
		PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓
DCP	TPAMI2010	13.45	0.701	0.04452	9.780	0.592	0.10520	10.89	0.572	0.08147	11.37	0.621	0.07706
FFA	AAAI2020	23.75	0.903	0.00422	26.50	0.941	0.00224	22.03	0.840	0.00627	24.09	0.894	0.00424
DCI-Net	TGRS2022	24.47	0.882	0.00357	25.05	0.910	0.00313	22.73	0.806	0.00533	24.08	0.866	0.00401
Restormer	CVPR2022	23.08	0.912	0.00492	24.73	0.903	0.00337	19.58	0.792	0.01102	22.46	0.869	0.00644
FSDGN	ECCV2022	26.18	0.914	0.00241	27.68	0.943	0.00171	23.95	0.854	0.00403	25.94	0.904	0.00271
Trinity-Net	TGRS2023	22.66	0.815	0.00542	25.02	0.929	0.00315	22.05	0.825	0.00624	23.24	0.856	0.00494
PSMB-Net	TGRS2023	26.75	0.928	0.00211	27.48	0.946	0.00179	25.15	0.889	0.00305	26.46	0.921	0.00232
C^2PNet	CVPR2023	19.62	0.880	0.01091	24.79	0.939	0.00332	16.83	0.790	0.02075	20.41	0.870	0.01166
FSNet	TPAMI2024	27.33	0.929	0.00185	26.95	0.940	0.00202	25.20	0.876	0.00302	26.49	0.915	0.00230
DEA-Net	TIP2024	27.11	0.931	0.00195	27.74	0.947	0.00168	24.82	0.877	0.00330	26.56	0.918	0.00231
MIMO	TGRS2024	27.36	0.929	0.00184	28.37	0.957	0.00146	22.83	0.851	0.00521	26.19	0.913	0.00284
DDN-SFF	PR2024	26.94	0.926	0.00203	29.73	0.943	0.00169	24.68	0.873	0.00340	26.45	0.914	0.00237
SFSNiD	CVPR2024	25.90	0.918	0.00257	26.58	0.937	0.00220	25.35	0.868	0.00367	25.61	0.908	0.00281
WinScaleFormer-L	JSTARS2025	27.80	0.979	0.00166	25.37	0.966	0.00290	23.29	0.938	0.00469	25.36	0.962	0.00291
DFG-DDM-Lite	TGRS2025	29.52	0.961	0.00112	28.82	0.957	0.00131	25.57	0.891	0.00277	27.97	0.936	0.00160
DS-RDMPD	TGRS2025	25.97	0.921	0.00251	28.30	0.952	0.00148	25.66	0.899	0.00273	26.94	0.924	0.00202
MABDT	Elsevier2025	27.52	0.948	0.00178	27.90	0.957	0.00161	24.07	0.868	0.00389	26.49	0.924	0.00242
SFRDP-Net	TGRS2025	27.91	0.928	0.00160	29.60	0.950	0.00106	25.93	0.881	0.00256	27.81	0.919	0.00174
DSCH-Net (Ours)	–	31.37	0.979	0.00072	31.40	0.981	0.00072	27.99	0.947	0.00158	30.25	0.969	0.00094

where I_{DH} and I_{GT} denote the dehazed output and ground-truth (GT) image, respectively, and N is the number of pixels in a mini batch.

To further enhance perceptual quality and structural consistency, a perceptual loss [61] is employed to measure the discrepancy between deep feature representations of the dehazed and GT images. In particular, features extracted from multiple intermediate layers of a pretrained VGG-16 network are

$$\mathcal{L}_P = \sum_{i=1}^{3} \frac{1}{c_i m_i n_i} \left\| V_i\left(I_{DH}\right) - V_i\left(I_{GT}\right) \right\|_2^2 \tag{19}$$

where $V_i(\cdot)$ represents the feature maps obtained from the ith selected layer, and c_i, m_i, and n_i correspond to the channel number, height, and width of the feature map, respectively.

The final training objective is obtained by linearly combining the two loss terms as

$$\mathcal{L}_T = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_P \tag{20}$$

where λ_1 and λ_2 are weighting coefficients that balance pixel accuracy and perceptual quality. In our implementation, these weights are set to $\lambda_1 = 1$ and $\lambda_2 = 0.04$, which provides a good tradeoff between convergence stability and visual performance.

G. Complexity

Let $N = HW$. Convolutions with small kernels are $O(Ck^2N)$, and all state-space scans in MD-SSB are $O(CN)$ per direction and run in parallel. PDE-DB uses fixed difference kernels and pointwise gates, also $O(CN)$, and RMDB adds only a constant number of dilated convolutions. Therefore, each stage is linear in image size, and DSCH-Net scales to high-resolution remote-sensing imagery without the quadratic overhead.

IV. EXPERIMENTAL RESULTS

A. Experimental Settings

1) Datasets: Evaluation is conducted on four publicly available remote-sensing dehazing benchmarks: RSID, RICE, SateHaze1K, and DHID. RSID contains approximately 1000 paired hazy/clean images at 256 × 256 resolution; following common practice, 900 pairs are used for training and 100 for testing. RICE includes two subsets: RICE1 (500 Google Earth pairs at 512 × 512) and RICE2 (450 Landsat-8 pairs). Each subset follows a 90/10 train-test protocol. SateHaze1K provides synthetic remote-sensing imagery under three haze levels: thin, moderate, and thick, with 400 pairs per level at 512 × 512. Adopt the standard split of 320 training pairs per level, with the remainder divided across validation and test. DHID is a large-scale dense-haze benchmark comprising 14 990 remote-sensing images generated by injecting heavy atmospheric haze into existing aerial scenes. In line with recent literature, 14 490 images are used for training, and 500 for testing [60]. Together, these datasets span synthetic and real/hybrid imagery, cover a wide range of haze densities, and offer diverse resolutions and scene structures, enabling a comprehensive assessment of dehazing performance.

2) Implementation Settings: DSCH-Net is implemented in PyTorch and trained end-to-end on a single NVIDIA RTX 4090 GPU. Unless otherwise stated, all experiments use a batch size of 16 and run for 200 epochs. Network parameters are optimized with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and an initial learning rate of 1×10^{-4}, reduced to 1×10^{-6} via cosine annealing. To increase robustness and prevent overfitting, standard geometric augmentations are applied on-the-fly, including random rotations of 90°, 180°, and 270°, as well as horizontal flips with probability 0.5. All datasets share identical hyperparameters and augmentation settings to ensure consistent and fair comparison across benchmarks.

3) Evaluation Metrics: Quantitative evaluation is conducted using peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and MSE. Higher PSNR and SSIM values indicate greater perceptual and structural fidelity, whereas lower MSE indicates more accurate pixel-level reconstruction.

TABLE II

COMPARISON RESULTS OF DSCH-NET WITH STATE-OF-THE-ART MODELS ON THE RICE AND DHID DATASETS, INCLUDING PARAMETERS. ↑: LARGER IS BETTER. ↓: SMALLER IS BETTER

Method	Venue/Year	RICE1			RICE2			DHID			#Param↓	FLOPs↓
		PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓		
DCP	TPAMI2010	12.39	0.628	0.05768	12.47	0.535	0.05662	12.96	0.517	0.05058	–	0.60G
FFA	AAAI2020	35.01	0.951	0.00032	34.82	0.894	0.00033	27.07	0.863	0.00196	4.69M	287.53G
DCI-Net	TGRS2022	35.03	0.960	0.00031	32.14	0.886	0.00061	28.17	0.892	0.00152	26.51M	26.86G
Restormer	CVPR2022	27.92	0.931	0.00161	24.61	0.801	0.00346	25.61	0.853	0.00275	26.10M	141.0G
FSDGN	ECCV2022	33.09	0.913	0.00049	33.56	0.893	0.00044	27.85	0.876	0.00164	2.73M	19.60G
Trinity-Net	TGRS2023	27.30	0.913	0.00186	24.71	0.794	0.00338	26.97	0.871	0.00201	20.14M	30.78G
PSMB-Net	TGRS2023	34.68	0.963	0.00034	33.96	0.893	0.00040	28.01	0.888	0.00158	12.49M	98.19G
C^2PNet	CVPR2023	28.64	0.911	0.00137	25.46	0.811	0.00284	27.85	0.881	0.00164	7.17M	460.95G
FSNet	TPAMI2024	37.06	0.965	0.00020	34.83	0.907	0.00033	28.51	0.907	0.00141	13.28M	111.14G
DEA-Net	TIP2024	36.12	0.963	0.00024	34.13	0.889	0.00039	28.12	0.906	0.00154	3.65M	32.23G
MIMO	TGRS2024	36.01	0.962	0.00025	34.28	0.904	0.00037	28.65	0.889	0.00136	15.28M	46.44G
DDN-SFF	PR2024	35.41	0.944	0.00029	34.64	0.903	0.00034	28.09	0.894	0.00155	29.5M	65.15G
SFSNiD	CVPR2024	35.53	0.965	0.00026	35.65	0.908	0.00034	28.12	0.903	0.00154	8.35M	69.36G
WinScaleFormer-L	JSTARS2025	36.65	0.985	0.00021	37.53	0.979	0.00017	28.37	0.891	0.00145	46.9M	121.87G
DFG-DDM-Lite	TGRS2025	37.01	0.959	0.00020	36.22	0.936	0.00024	27.51	0.852	0.00178	33.7M	347.12G
DS-RDMPD	TGRS2025	36.81	0.968	0.00017	36.64	0.945	0.00021	27.80	0.880	0.00166	10.85M	272.6G
MABDT	Elsevier2025	32.45	0.946	0.00057	33.10	0.911	0.00049	26.08	0.860	0.00245	12.06M	141.7G
SFRDP-Net	TGRS2025	37.30	0.962	0.00019	35.41	0.927	0.00028	28.99	0.913	0.00124	5.12M	40.51G
DSCH-Net (Ours)	–	38.91	0.990	0.00017	39.50	0.959	0.00011	29.56	0.918	0.00110	1.66M	24.22G

B. Quantitative Analysis

The proposed DSCH-Net demonstrates consistently strong performance across all benchmarks, as shown in Tables I and II, reflecting the combined effects of physics-guided diffusion, multidirectional state-space modeling, and multidilation texture recovery. On the SateHaze1K dataset, which includes thin, moderate, and thick haze levels, the model achieves noticeably higher PSNR and SSIM scores than prior CNN, Transformer, diffusion, and frequency-domain approaches. The improvements are most evident under thin and moderate haze, where accurate color restoration and structure preservation are crucial. Even under thick-haze conditions, where high haze density remains challenging, DSCH-Net maintains a clear margin over the most recent TGRS and CVPR methods.

The evaluation on RICE1 and RICE2 further highlights the model's generalization ability across different remote-sensing sources. Both Google Earth and Landsat-8 imagery exhibit large variations in color tone, ground texture, and acquisition conditions, yet DSCH-Net consistently delivers the highest reconstruction quality among all competing methods.

Performance on the dense-haze DHID dataset reinforces these observations. This benchmark comprises highly degraded aerial images with strong, spatially heterogeneous haze. Despite the difficulty, DSCH-Net provides measurable improvements over recent state-of-the-art techniques, recovering sharper boundaries and more reliable structural cues in vegetation, coastlines, and urban regions. The stability of the physics-guided diffusion update reduces haze veil thickness, while selective skip fusion helps avoid haloing at bright sky–object transitions.

The baseline results reported in Tables I and II are obtained from two sources: 1) local reimplementation and retraining under identical experimental settings for methods with available and compatible implementations (including SFRDP-Net, MABDT, DS-RDMPD, and DFG-DDM-Lite) and 2) direct citation from original publications for earlier state-of-the-art methods whose implementations are unavailable or not applicable to the evaluated dataset.

C. Qualitative Analysis

To provide additional insight beyond quantitative metrics, a comprehensive qualitative evaluation is conducted across diverse remote-sensing scenarios. The comparisons in Fig. 3 highlight how different models handle structural fidelity, color consistency, and haze suppression under varying levels of atmospheric degradation.

In the first two sets of examples, competing methods such as SFSNiD [62], WinScaleFormer-L [63], DFG-DDM-Lite [64], DS-RDMPD [65], MABDT [66], and SFRDP-Net [36] partially reduce haze but often exhibit residual artifacts, oversmoothing, or inaccurate color restoration. In contrast, DSCH-Net reconstructs sharper edges, recovers finer textures, and preserves natural color tones while avoiding common reconstruction failures such as color shifting or halo formation. These improvements arise from the synergy between the multidirectional state-space interactions, physics-guided diffusion updates, and residual multidilation texture refinement.

The corresponding pixelwise error maps (middle rows of Fig. 3) further emphasize the robustness of the proposed approach. Competing methods display dense, high-error regions, particularly around man-made structures and textured areas. DSCH-Net reduces both localized and global reconstruction errors, yielding cleaner, more uniform error distributions. Normalized error visualizations (bottom rows) confirm that the proposed model generalizes well across scene types, maintaining structural coherence even in heavily degraded regions.

Additional examples from the RICE and DHID datasets in Fig. 4 reinforce these observations. While other models often leave noticeable haze or distort textures, DSCH-Net consistently produces clearer outputs with minimal residual error, demonstrating strong generalization across different scenes.

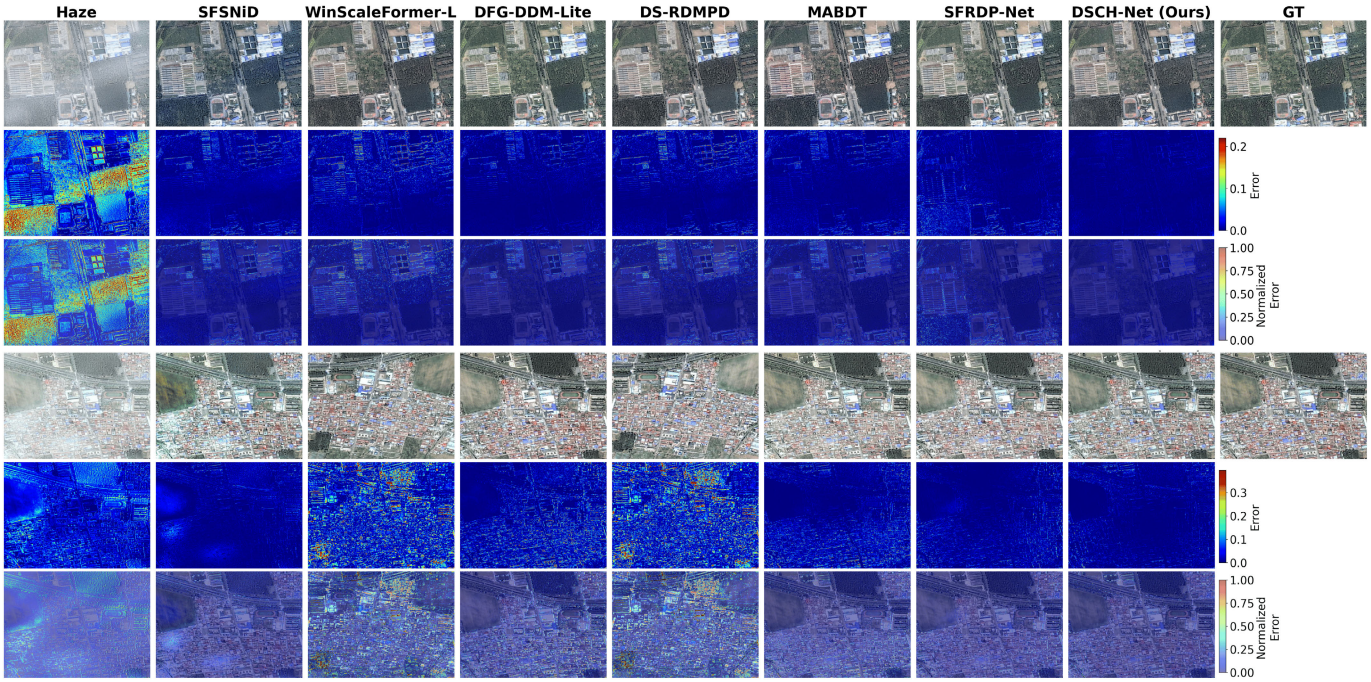


Fig. 3. Qualitative comparison of different dehazing methods on remote sensing imagery. Row 1 presents the dehazed outputs of each model, along with the GT. Row 2 provides the corresponding pixelwise error maps [mean squared error (mse)] computed against GT using a unified color scale, enabling consistent cross-method comparison. Row 3 visualizes the normalized error overlaid on each prediction, revealing the spatial distribution of residual haze, color inconsistencies, and structural distortions.

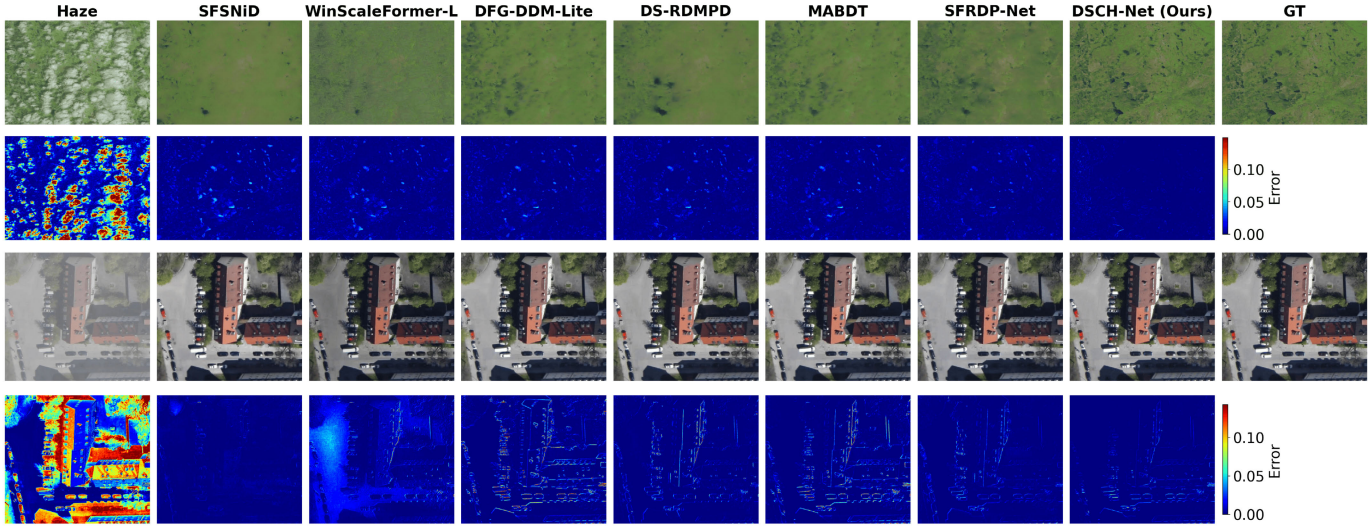


Fig. 4. Visual evaluation of different dehazing approaches on the RICE and DHID datasets. The top row shows the estimated haze-free outputs from competing methods, and the bottom row depicts the corresponding error maps, computed as per-pixel mean-squared differences.

D. Ablation Analysis

To assess the contribution of each major component in DSCH-Net, comprehensive ablation experiments are conducted on the RSID dataset. The analysis is divided into three parts: 1) core architectural modules; 2) directional scanning strategies in the MD-SSB block; and 3) PDE diffusion hyperparameters. Each subsection presents both quantitative results and a short interpretation.

Since RSID is newly introduced and not yet widely adopted, this article additionally provides an overall comparison with representative baseline and state-of-the-art dehazing methods to contextualize absolute performance. Table III summarizes these results. For models with publicly available implementations and feasible training pipelines (including SFRDP-Net, MABDT, DS-RDMPD, and DFG-DDM-Lite), local reimplementation and retraining were conducted under identical experimental settings to ensure fairness. For other methods whose official implementations are unavailable or incompatible with RSID, the reported results are directly cited from the original publications. This combined evaluation enables fair componentwise analysis while providing a meaningful absolute performance reference.

1) Effect of Core Architectural Components: Table IV and the visual evidence in Fig. 5 report the progressive integration of the central blocks: PDE-DB, MD-SSB, and RMDB. Starting

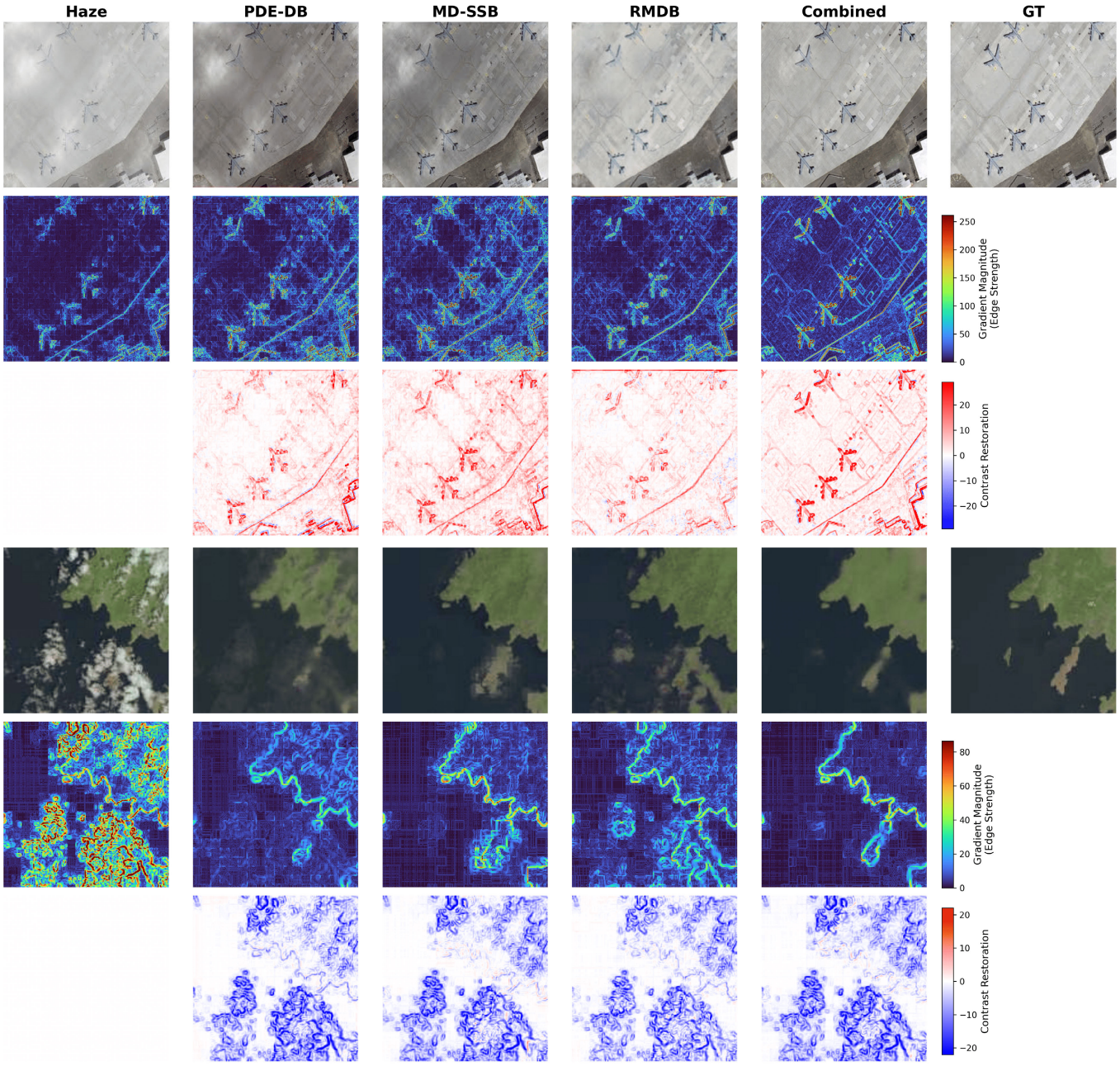


Fig. 5. Ablation study visualization. For each patch, the first row presents the hazy input, the outputs obtained from individual architectural components (PDE-DB, MD-SSB, and RMDB), the full combined variant of the proposed network, and the corresponding GT. The second row visualizes the gradient-magnitude responses, which highlight structural fidelity and edge recovery across the evaluated configurations. The third row illustrates the contrast-restoration maps, computed as the local standard deviation gain relative to the hazy input, thereby revealing the contribution of each component to local contrast enhancement.

from a minimal encoder–decoder baseline, adding the PDE block already enhances both PSNR and SSIM, confirming that physics-guided diffusion effectively suppresses low-frequency haze veil while retaining important structures. Embedding the MD-SSB module further improves global contextual propagation performance through directional state-space scans.

2) Sensitivity Analysis of Dilation Rates in RMDB: To further analyze the sensitivity of RMDB to dilation-rate selection, a focused ablation study was conducted. The dilation candidates were selected to represent three complementary receptive-field scales $d = 1$ for local texture restoration, $d = 4$ for mid-range spatial variation, and $d = 8$ for broader contextual support. To analyze sensitivity without an exhaustive search over many dilation values, all pairwise subsets and the full triple formed from these representative rates, i.e., $(1, 4)$, $(1, 8)$, $(4, 8)$, and $(1, 4, 8)$, were evaluated, as results are shown in Table IV. The results show a consistent improvement when moving from varied two-rate settings to the three-rate setting, indicating that omitting either the local or the broader dilation reduces the ability to recover fine structures and maintain large-area consistency under nonuniform haze. Therefore, $(1, 4, 8)$ is adopted as a principled multiscale configuration with robust performance.

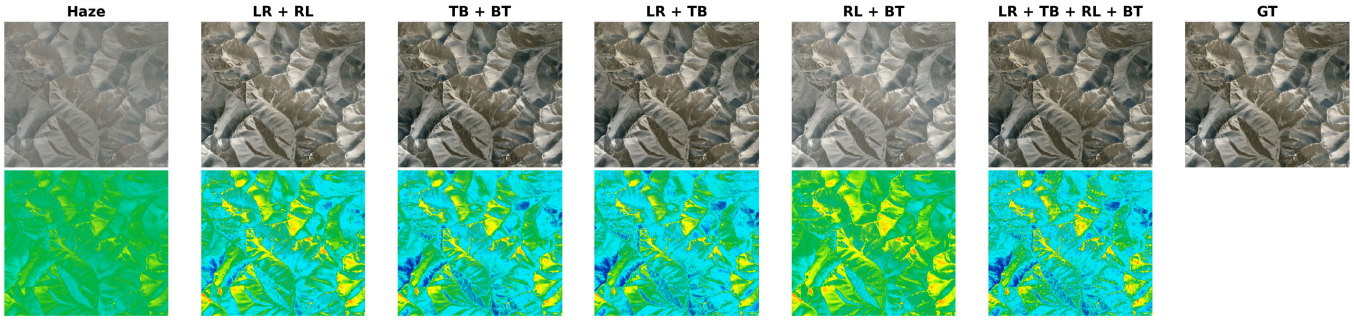


Fig. 6. Ablation comparison in directional scanning. The first row presents the restored images produced using paired scanning directions (LR+RL, TB+BT, LR+TB, RL+BT) and the full four-direction configuration (LR+TB+RL+BT). In contrast, the second row shows the corresponding thermal-style enhancement maps for each configuration, highlighting differences in recovered structural cues and local contrast. The full directional combination yields the most consistent enhancement across terrain features, demonstrating the complementary contribution of multidirectional scanning in the reconstruction process.

TABLE III

QUANTITATIVE COMPARISON ON THE RSID DATASET WITH STATE-OF-THE-ART MODELS

Method	Venue/Year	PSNRR↑ (dB)	SSIMR↑	MSER↓
DCP	TPAMI2010	13.33	0.7021	0.04643
DCI-Net	TGRS2022	19.81	0.7630	0.01045
FFA-Net	AAAI2020	24.05	0.8993	0.00394
FSDGN	ECCV2022	25.38	0.9430	0.00290
Trinity-Net	TGRS2023	24.05	0.8990	0.00394
PSMB-Net	TGRS2023	25.20	0.9440	0.00302
C^2PNet	CVPR2023	24.12	0.9350	0.00387
FSNet	TPAMI2024	28.40	0.9520	0.00145
DEA-Net	TIP2024	26.15	0.9440	0.00243
DFG-DDM-Lite	TGRS2025	28.43	0.9615	0.00144
DS-RDMPD	TGRS2025	26.29	0.9532	0.00240
MABDT	Elsevier2025	27.94	0.9608	0.00161
SFRDP-Net	TGRS2025	29.87	0.9710	0.00103
DSCH-Net (Ours)	–	32.20	0.9800	0.00060

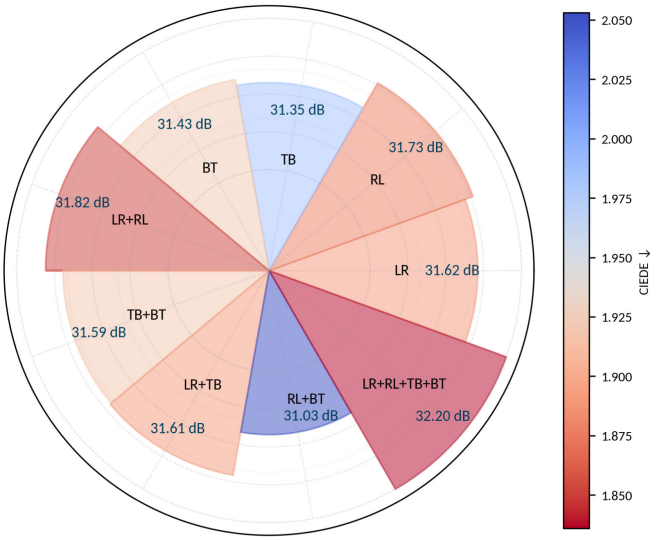


Fig. 7. Ablation of directional scanning strategies in the MD-SSB module. Each petal indicates a specific directional configuration (LR, RL, TB, BT, and combinations), with radius representing PSNR and color representing CIEDE.

3) Impact of Directional Scanning in MD-SSB: Directional scanning is at the heart of MD-SSB. Table V, Fig. 6, and the visual comparison in Fig. 7 jointly examine the effect of individual and combined scanning directions. Single-direction scans (LR, RL, TB, and BT) already yield noticeable improvements, as each direction captures a different subset of

TABLE IV

ABLATION OF CORE ARCHITECTURAL COMPONENTS ON THE RSID DATASET

Method Variant	PSNR↑	SSIM↑	CIEDE↓	MSE↓
Baseline	29.87	0.971	2.401	0.001034
+ PDE Block	30.48	0.976	2.102	0.001487
+ PDE + MD-SSB	31.28	0.979	1.962	0.001110
+ PDE + MD-SSB + RMDB	**32.20**	**0.980**	**1.836**	**0.000847**
+ RMDB (1,4)	31.29	0.978	2.061	0.001101
+ RMDB (1,8)	31.97	0.979	1.855	0.000878
+ RMDB (4,8)	32.00	0.980	1.845	0.000862
+ RMDB (1,4,8)	32.20	0.980	1.836	0.000847

TABLE V

EFFECT OF DIRECTIONAL SCANNING STRATEGIES ON THE RSID DATASET

Scan Direction	PSNR↑	SSIM↑	CIEDE↓	MSE↓
LR	31.62	0.979	1.890	0.000900
RL	31.73	0.979	1.877	0.000891
TB	31.35	0.978	1.985	0.000980
BT	31.43	0.979	1.920	0.000920
LR + RL	31.82	0.979	1.851	0.000885
TB + BT	31.59	0.978	1.919	0.000881
LR + TB	31.61	0.979	1.891	0.000899
RL + BT	31.03	0.978	2.053	0.001077
LR + RL + TB + BT	**32.20**	**0.980**	**1.836**	**0.000847**

structural cues. Pairwise combinations further strengthen these cues, with horizontal (LR+RL) and vertical (TB+BT) pairings offering complementary benefits. The qualitative evidence in Fig. 6 shows that richer texture recovery and more stable local contrast emerge when directional paths are combined. The best performance, both numerically and visually, is achieved when all four directions operate simultaneously, confirming that full multidirectional contextual modeling is essential for producing stable, coherent, and artifact-free reconstructions.

4) Influence of PDE-DB Hyperparameters and Comparative Design Analysis: The PDE-DB is designed to suppress the low-frequency haze veil while preserving structural boundaries under spatially nonuniform haze. Its effectiveness depends not only on internal hyperparameters but also on the diffusion formulation itself. This subsection analyzes both aspects.

a) Effect of diffusion hyperparameters: Table VI investigates two internal factors of the PDE-DB: the diffusion step size and the presence of the learnable gating mechanism.

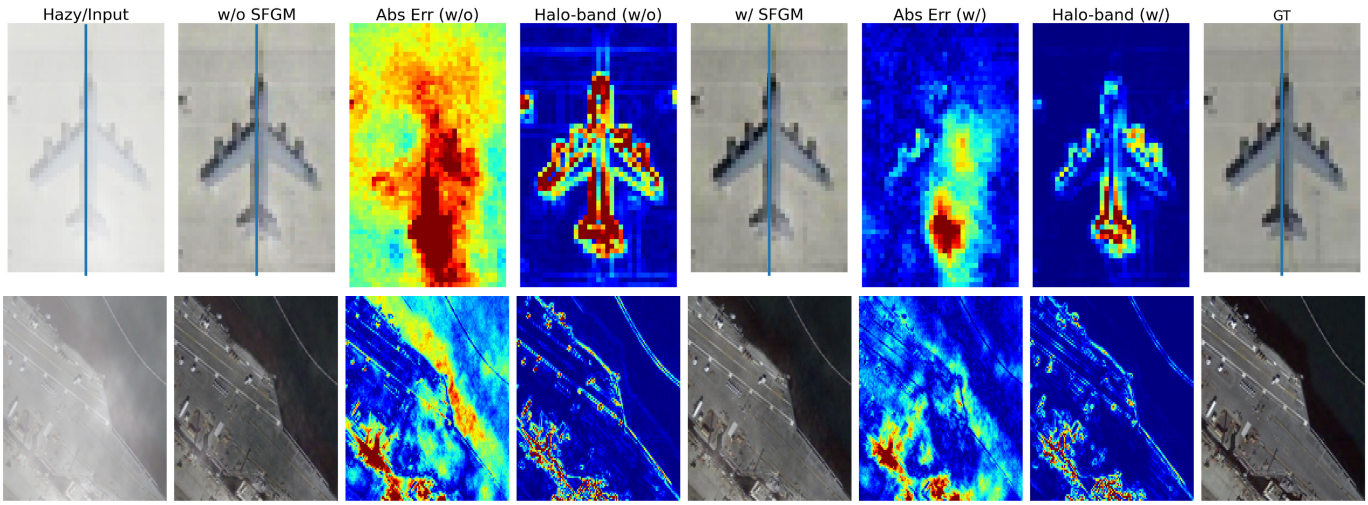


Fig. 8. Effect of SFGM on halo artifacts. From left to right: hazy input, dehazed result without SFGM, absolute error and halo-band maps (without SFGM), dehazed result with SFGM, absolute error and halo-band maps (with SFGM), and GT. SFGM effectively suppresses halo artifacts and reduces intensity overshoot near sharp transitions.

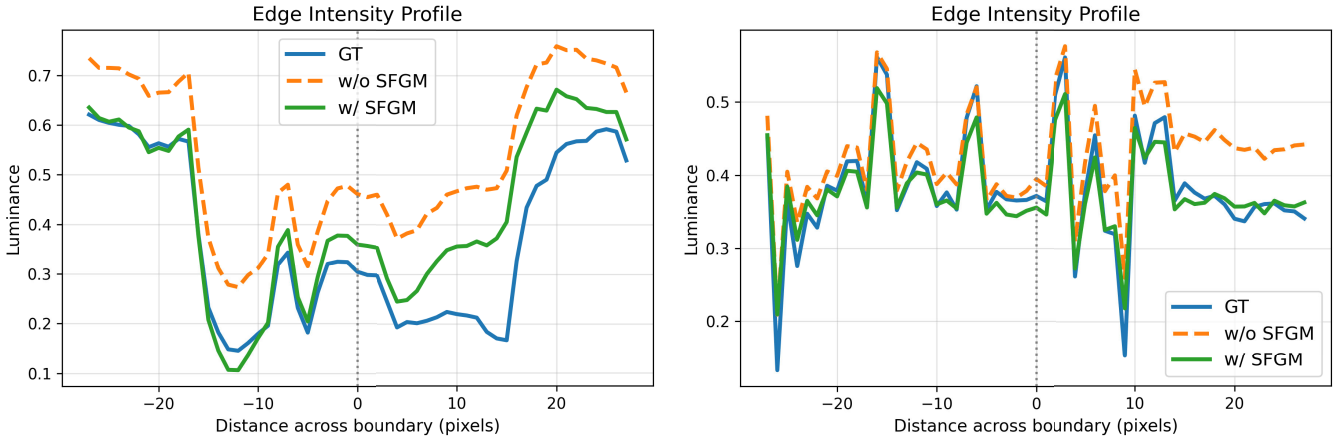


Fig. 9. Luminance intensity profiles sampled across a representative object boundary. Compared to naive skip fusion (without SFGM), the proposed SFGM noticeably reduces intensity overshoot and postedge oscillations (halo artifacts), producing smoother edge transitions that more closely follow the GT profile.

TABLE VI
INFLUENCE OF PDE HYPERPARAMETERS ON THE RSID DATASET

Configuration	PSNR↑	SSIM↑	CIEDE↓	MSE↓
Step size = 0.05	**32.20**	**0.980**	**1.802**	**0.000833**
Step size = 0.1	32.20	0.980	1.836	0.000847
Step size = 0.2	31.96	0.979	1.849	0.000853
Gate = True	**32.20**	**0.980**	**1.836**	**0.000847**
Gate = False	30.98	0.978	2.053	0.001133

From a diffusion-theoretic perspective, moderate step sizes (0.05–0.1) yield the most stable reconstructions by sufficiently attenuating low-frequency haze components while avoiding excessive smoothing. This behavior is consistent with classical anisotropic diffusion, where controlled update magnitudes prevent the destruction of high-frequency structures. In contrast, larger step sizes lead to over-smoothing, resulting in degradation of thin structures and local contrast.

The learnable gating mechanism has a substantially greater impact than step-size tuning. Disabling the gate causes a clear and consistent drop in PSNR, SSIM, and color fidelity, indicating that uniform diffusion is inadequate for remote-sensing imagery with spatially varying haze density. Unlike traditional physics-inspired dehazing methods [67], [68], which rely on fixed transmission or atmospheric priors, the gating mechanism adaptively modulates diffusion strength based on local feature statistics. This enables selective smoothing in homogeneous regions while suppressing diffusion near strong edges and fine structures.

b) Comparison with alternative feature enhancement designs: To isolate the unique contribution of the PDE formulation beyond hyperparameter tuning, additional controlled ablation experiments replace the PDE-DB with commonly used feature enhancement alternatives while keeping the rest of the architecture unchanged. The following variants are evaluated.

1) *No PDE*: The diffusion block is removed entirely.
2) *Residual Conv*: The PDE-DB is replaced with a standard residual block composed of two 3×3 convolutions.

TABLE VII
COMPARISON OF PDE-DB WITH ALTERNATIVE FEATURE ENHANCEMENT DESIGNS ON RSID

Variant	PSNR↑	SSIM↑	CIEDE↓	MSE↓	Params (M)↓	FLOPs (G)↓
No PDE	30.98	0.978	2.053	0.001133	1.52	22.1
Residual Conv	31.24	0.976	1.982	0.001020	1.59	23.4
Residual + SE	31.53	0.977	1.901	0.000940	1.70	24.9
PDE-DB (proposed)	**32.20**	**0.980**	**1.836**	**0.000847**	1.66	24.2

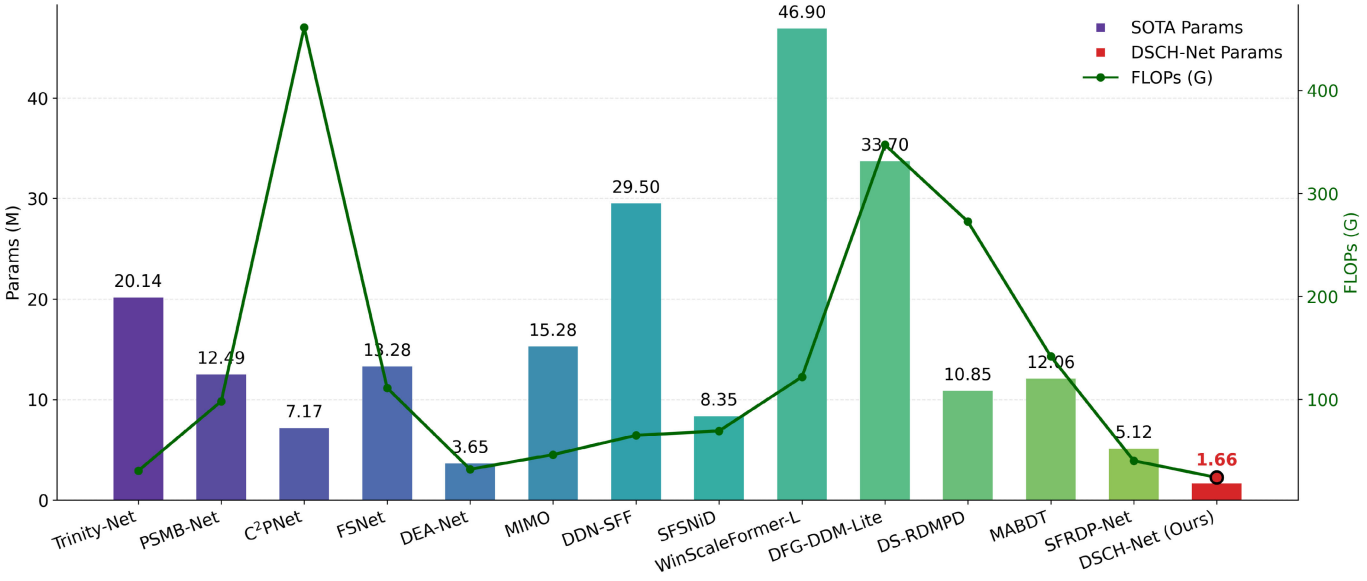


Fig. 10. Comparative analysis of model complexity across recent dehazing networks. Bar heights denote the number of parameters (M), while the green curve represents computational cost measured in FLOPs (G). DSCH-Net achieves the lowest parameter count and significantly fewer FLOPs than existing state-of-the-art approaches.

3) *Residual + Channel Attention*: A squeeze-and-excitation module is added to the residual block.
4) *PDE-DB (Proposed)*: The full physics-guided diffusion formulation.

All variants are trained under identical settings on the RSID dataset. Both reconstruction accuracy and computational overhead are reported. Standard residual blocks improve performance relative to removing the diffusion mechanism but remain limited to local aggregation and lack explicit modeling of the low-frequency haze veil. Channel-attention-based variants further enhance selectivity but introduce additional parameters and computational cost without addressing spatially adaptive haze suppression, as shown in Table VII.

In contrast, the PDE-DB consistently achieves the best reconstruction accuracy with moderate computational overhead by explicitly embedding edge-aware diffusion behavior into the feature update. This confirms that the performance gains are from the physics-guided diffusion formulation rather than increased network capacity. Overall, these results demonstrate that spatially adaptive, physics-guided diffusion is essential for handling nonuniform haze distributions in complex aerial imagery. The PDE-DB provides a stable and physically grounded representation that complements the subsequent state-space contextual modeling and multidilation texture refinement.

5) Effect of Selective Fusion Gating: To explicitly analyze the role of the SFGM, a comparison is performed between the SFGM and the case without SFGM under identical training and inference settings. Fig. 8 visualizes representative boundary regions together with absolute error maps and halo-band maps. For each test image, boundary regions are manually selected around high-contrast transitions (e.g., object–sky interfaces). Absolute error maps are computed as the pixelwise difference between the dehazed output and the GT. To further highlight halo-related distortions near edges, we define a halo-band map as the product of the absolute error and the gradient magnitude of the GT luminance image. This formulation emphasizes reconstruction errors concentrated along strong structural boundaries. In addition, edge intensity profiles sampled across object boundaries as presented in Fig. 9 show that SFGM produces luminance transitions that more closely follow the GT profile, whereas the ungated variant exhibits exaggerated peaks and ringing.

6) Computational Complexity Study: The computational footprint of DSCH-Net is analyzed in comparison with recent state-of-the-art dehazing models. Table II reports the parameter counts and FLOP requirements of competing architectures, while Fig. 10 visualizes the corresponding complexity trends. Most contemporary dehazing networks rely on deep CNN backbones or attention-heavy architectures, resulting in parameter budgets of 10–25 M and computational costs of 60–100 G FLOPs. In contrast, DSCH-Net adopts a sig-

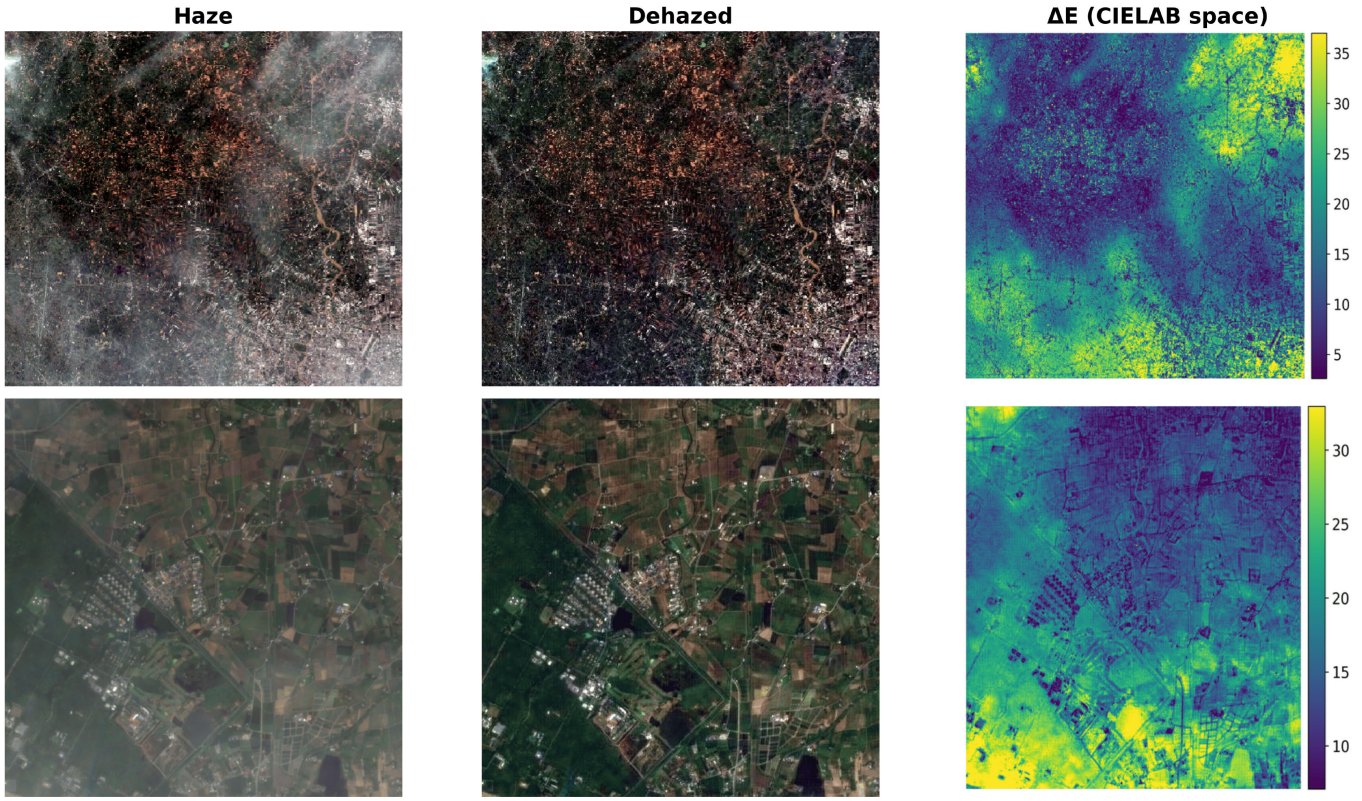


Fig. 11. Real Sentinel-2 Level-2A RGB composite over Bangkok. (Left) Original hazy image. (Middle) DSCH-Net output. (Right) Pixelwise ΔE map in CIELAB space highlighting spatial distribution of perceptual color correction. Higher ΔE values are concentrated in haze-dense regions, while structurally clear areas exhibit lower deviation, indicating physically consistent atmospheric attenuation without random chromatic artifacts.

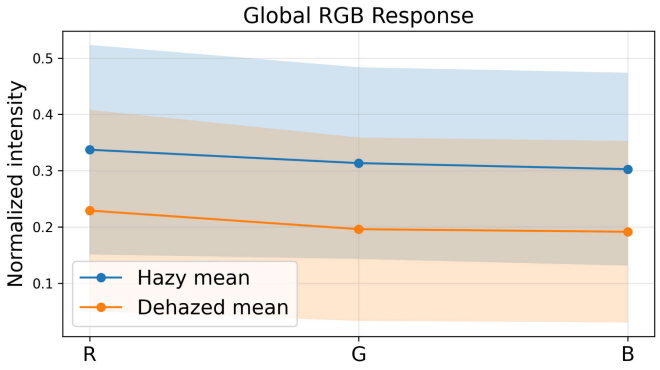


Fig. 12. Global visible-band radiometric response before and after dehazing on Sentinel-2 imagery. Mean intensity decreases consistently across R, G, and B channels following DSCH-Net processing, reflecting attenuation of additive atmospheric airlight. The preserved spectral ordering $R>G>B$ indicates stable wavelength-dependent reflectance relationships. Shaded regions denote standard deviation.

nificantly more compact configuration with only 1.66 M parameters and 24.22 G FLOPs, making it the lightest model among the compared methods [58], [59], [62], [63], [64], [65], [66]. Despite this substantial reduction in complexity, DSCH-Net consistently achieves state-of-the-art performance across all evaluated benchmarks.

This favorable accuracy–efficiency tradeoff arises from the coordinated design of diffusion-based modeling and state-space global context propagation. In particular, the MD-SSB replaces quadratic self-attention with four axial state-space scans whose complexity scales linearly with image resolution, enabling global dependency modeling without expensive token-to-token interactions. In addition, the overall architecture avoids excessive channel expansion by employing lightweight operators, including depthwise and 1×1 pointwise convolutions for gating and projection, diagonal or channelwise state-space parameters in the recurrence, and a small number of parallel dilated paths in RMDB. The PDE-DB further incorporates fixed finite-difference stencils with learnable step sizes and conductance maps, injecting physics-inspired behavior without introducing deep convolutional stacks. Compared to prior dehazing approaches [39], [45], [46], [63], [64], [65] that combine physics-guided formulations with heavy CNN backbones, the proposed design enables efficient feature propagation while maintaining strong reconstruction capability.

E. Real-World Evaluation on Sentinel-2 Imagery

To evaluate cross-sensor robustness under real atmospheric conditions, additional experiments were conducted on Sentinel-2 Level-2A surface reflectance imagery acquired over Bangkok with dimensions of 573×511 pixels extracted from a 100×100 Sentinel-2 tile covering approximately 28 km^2 of the study region. Recent studies have also explored large-scale remote-sensing image processing using foundation models and large-scene analysis frameworks [69], [70], highlighting the growing importance of handling large-area satellite imagery in practical applications. The model trained

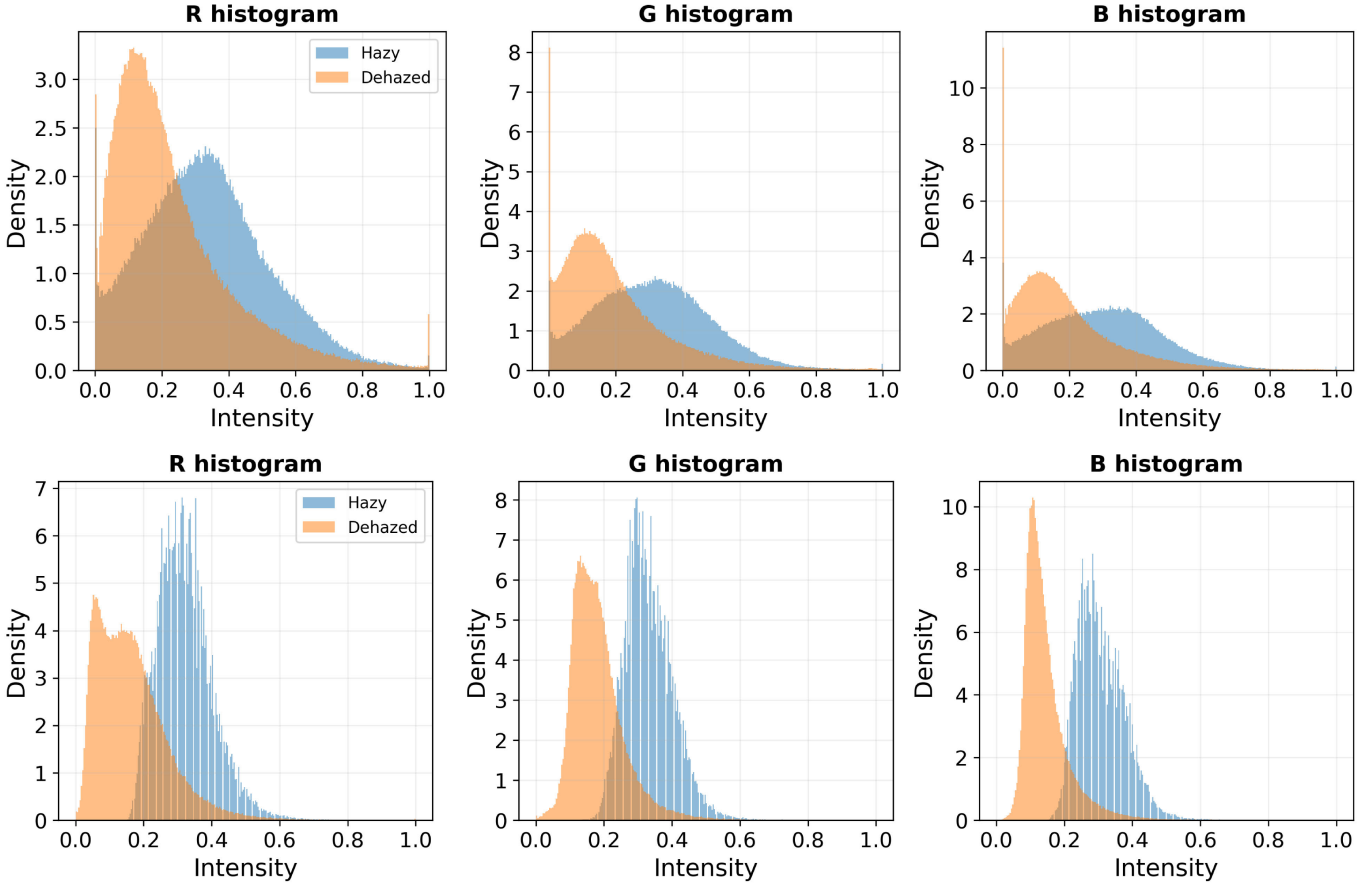


Fig. 13. Channelwise intensity distributions of Sentinel-2 imagery before and after dehazing. All visible bands exhibit a leftward shift after DSCH-Net processing, consistent with atmospheric light removal. The distributions remain smooth and unimodal without saturation spikes or multimodal distortion, indicating stable radiometric redistribution.

on RSID was applied to assess generalization across different sensor spectral-response characteristics. Fig. 11 presents the original hazy RGB composite, the DSCH-Net output, and the corresponding pixelwise ΔE map in CIELAB space.

Visually, DSCH-Net suppresses haze and thin cloud veils while preserving urban structures, road networks, and river boundaries without over-enhancement. The ΔE map provides a spatial quantification of perceptual color change between hazy and dehazed images. Rather than exhibiting random chromatic distortions, the corrections are spatially structured and concentrated in haze-dense regions. Clear surface areas show comparatively lower deviation. The global statistics (mean 16.38, median 15.11, standard deviation 9.05, 95th percentile 32.75) indicate controlled chromatic adjustment driven by physically meaningful haze attenuation.

1) Visible-Band Radiometric Consistency: Because atmospheric haze primarily introduces an additive airlight component that increases brightness in visible wavelengths, the global channel statistics were used to verify whether the correction behaves consistently across spectral bands.

As illustrated in Fig. 12, the global mean intensities decrease across R, G, and B channels following dehazing, with reductions of 32.05%, 37.46%, and 36.75%, respectively. This consistent attenuation indicates removal of additive atmospheric scattering rather than arbitrary brightness suppression. Importantly, the relative spectral ordering $R > G > B$ remains preserved before and after correction, suggesting that intrinsic wavelength-dependent reflectance relationships are maintained. Channel variance changes moderately, which indicates controlled radiometric redistribution without excessive contrast amplification or clipping artifacts.

2) Intensity Distribution Stability: To further assess whether the correction introduces distributional distortions, the channelwise intensity histograms are examined in Fig. 13. The histograms exhibit a consistent leftward shift after dehazing, reflecting attenuation of atmospheric airlight. Crucially, the distributions remain smooth and unimodal, without saturation spikes or multimodal artifacts. This confirms stable radiometric correction rather than aggressive histogram stretching or nonlinear intensity remapping. Overall, the combined visual, perceptual, and radiometric analyses demonstrate that DSCH-Net attenuates wavelength-dependent scattering effects while preserving the visible band. These results confirm the sensor's generalization stability on real Sentinel-2 imagery.

V. Conclusion

This work presents DSCH-Net, a physics-guided state-space network for single-image remote-sensing dehazing. Rather than treating dehazing as a purely data-driven mapping, the model is built around a mechanistic decomposition

of the degradation. The PDE-driven diffusion block targets low-frequency haze by performing a learnable, edge-aware smoothing process in feature space, which suppresses airlight-induced color shifts while preserving discontinuities via spatially adaptive conductance. The MD-SSB then propagates information across long spatial extents with linear complexity, acting as a global mixing operator that enforces large-area consistency in sky gradients and illumination while avoiding the quadratic interactions of attention. Finally, the RMDB restores high-frequency content by aggregating complementary receptive fields, recovering thin structures and textures that are typically attenuated after haze suppression. A selective fusion gate stabilizes encoder–decoder feature reuse by filtering incompatible skip signals, which reduces halo formation at strong depth and brightness transitions such as sky–object boundaries.

Extensive experiments on RSID, SateHaze1K, RICE1/2, and DHID demonstrate that this structured division of labor yields consistent improvements in PSNR, SSIM, MSE, and color fidelity, while maintaining a compact model size. The ablation results align well with the intended representational roles of each module. In particular, PDE-DB primarily enhances chromatic stability and removes low-frequency haze. MD-SSB improves global coherence and long-range structural consistency. RMDB focuses on recovering fine-grained details. When directional scanning or diffusion gating is removed, robustness under nonuniform haze is noticeably degraded. This degradation occurs because both global coupling and edge-aware diffusion are required to prevent residual haze veils and oversmoothing. Overall, DSCH-Net admits a clear mechanistic interpretation. Dehazing is achieved by jointly modeling low-frequency atmospheric scattering, enforcing global contextual consistency through efficient state-space propagation, and reconstructing high-frequency details via multiscale residual refinement. It is worth noting that, due to the absence of GT haze-free references in real remote-sensing scenarios, evaluation on real imagery remains primarily qualitative.

VI. Limitations and Future Work

Although DSCH-Net achieves strong quantitative and qualitative results, several limitations remain. First, most of the experiments rely on paired synthetic or semi-synthetic benchmarks. While DHID and RSID include realistic structures and strong haze, they still do not cover the full variability of real atmospheric conditions, sensors, and seasonal changes. The generalization behavior under extreme domain shifts (e.g., very low solar angles, snow, sandstorms, or mixed weather) remains underexplored.

Second, the current model processes each image independently and assumes a single RGB modality. Many operational remote-sensing systems provide multispectral, hyperspectral, SAR, or LiDAR data and often acquire temporal sequences. DSCH-Net does not yet exploit such complementary information, which could help disambiguate haze from natural low-contrast regions and improve temporal stability. In particular, mixed conditions such as haze combined with thin cirrus clouds can be difficult to disentangle from a single RGB observation, and incorporating multispectral/thermal cues or temporal sequences is a promising direction to better separate these atmospheric effects.

These limitations suggest several directions for future work. One is to design self-supervised or unpaired training schemes that leverage extensive collections of in-the-wild aerial and satellite images without clean references. Another is to extend DSCH-Net to multimodal and temporal settings, for example, by integrating depth, polarization, or SAR channels and modeling temporal consistency across image sequences. It is also promising to develop ultra-lightweight variants with hardware-aware design, including quantization and neural architecture search for edge platforms. Finally, plan to embed the proposed backbone into end-to-end pipelines for mapping, object detection, and environmental monitoring, and to evaluate its benefits under real operational constraints.

References

[1] J. Yang, Z. Wang, D. Chen, S. Xiao, and A. T. Azar, "A joint network of super-resolution and active learning for agricultural land classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13669–13677, 2024.

[2] A. Jabbar et al., "Deep transfer learning-based automated diabetic retinopathy detection using retinal fundus images in remote areas," *Int. J. Comput. Intell. Syst.*, vol. 17, no. 1, p. 135, May 2024.

[3] M. Zakzouk, I. A. El-Magd, E. M. Ali, A. M. Abdulaziz, A. Rehman, and T. Saba, "Novel oil spill indices for sentinel-2 imagery: A case study of natural seepage in Qaruh island, Kuwait," *MethodsX*, vol. 12, Jun. 2024, Art. no. 102520.

[4] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

[5] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.

[6] E. Akleman, "Deep learning," *Computer*, vol. 53, no. 9, pp. 1–17, Sep. 2020.

[7] Z. Zheng et al., "Ultra-high-definition image dehazing via multi-guided bilateral learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16180–16189.

[8] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016, doi: 10.1109/TPAMI.2015.2439281.

[9] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[10] X. Tao, H. Gao, X. Shen, J. Wang, and J. Jia, "Scale-recurrent network for deep image deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8174–8182.

[11] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, "Single image dehazing via multi-scale convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 154–169.

[12] D. Ren, W. Zuo, Q. Hu, P. Zhu, and D. Meng, "Progressive image deraining networks: A better and simpler baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3937–3946.

[13] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 105–114.

[14] M. S. M. Sajjadi, B. Scholkopf, and M. Hirsch, "EnhanceNet: Single image super-resolution through automated texture synthesis," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4491–4500.

[15] X. T. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Comput. Vis. ECCV Workshops*, vol. 11133, 2019, pp. 63–79.

[16] J. W. Soh, G. Y. Park, J. Jo, and N. I. Cho, "Natural and realistic single image super-resolution with explicit natural manifold discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8114–8123.

[17] W. Wang, R. Guo, Y. Tian, and W. Yang, "CFSNet: Toward a controllable feature space for image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4140–4149.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[19] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[20] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22367–22377.

[21] Z. Tu et al., "MaxViT: Multi-axis vision transformer," in *Proc. 17th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 459–479.

[22] H. Li, Y. Zhang, J. Liu, and Y. Ma, "GTMNet: A vision transformer with guided transmission map for single remote sensing image dehazing," *Sci. Rep.*, vol. 13, no. 1, p. 9222, Jun. 2023, doi: 10.1038/s41598-023-36149-6.

[23] B. Huang, Z. Li, C. Yang, F. Sun, and Y. Song, "Single satellite optical imagery dehazing using SAR image prior based on conditional generative adversarial networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1795–1802, doi: 10.1109/WACV45572.2020.9093318.

[24] W. Dong et al., "End-to-end detail-enhanced dehazing network for remote sensing images," *Remote Sens.*, vol. 16, no. 2, p. 225, Jan. 2024, doi: 10.3390/rs16020225.

[25] B. Benjdira, A. M. Ali, and A. Koubaa, "Streamlined global and local features combinator (SGLC) for high resolution image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1855–1864.

[26] S. Bianco, L. Celona, F. Piccoli, and R. Schettini, "High-resolution single image dehazing using encoder–decoder architecture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 1927–1935.

[27] X. Chen, Y. Li, L. Dai, and C. Kong, "Hybrid high-resolution learning for single remote sensing satellite image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3072917.

[28] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12294–12305, doi: 10.1109/CVPR46437.2021.01212.

[29] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844, doi: 10.1109/ICCVW54120.2021.00212.

[30] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5728–5739, doi: 10.1109/CVPR52688.2022.00565.

[31] Z. Wang, X. Cun, J. Bao, W. Zhou, J. Liu, and H. Li, "Uformer: A general U-shaped transformer for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17662–17672, doi: 10.1109/CVPR52688.2022.01718.

[32] C. Li, S. Zhou, T. Wu, J. Shi, and F. Guo, "A dehazing method for UAV remote sensing based on global and local feature collaboration," *Remote Sens.*, vol. 17, no. 10, p. 1688, 2025.

[33] A. M. Ali, E.-S.-M. El-Rabaie, K. F. Ramadan, F. E. A. El-Samie, and W. El-Shafai, "CLEAR-Net: A cascaded local and external attention network for enhanced dehazing of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, pp. 1–20, 2025, doi: 10.1109/JSTARS.2025.3534984.

[34] H. Sun et al., "Bidirectional-modulation frequency-heterogeneous network for remote sensing image dehazing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 11, pp. 10649–10664, Nov. 2025, doi: 10.1109/TCSVT.2025.3570998.

[35] M. Miao, H. Huang, K. Huang, and S. Wang, "A dual branch network combining detail information and color feature for remote sensing image dehazing," *Int. J. Mach. Learn. Cybern.*, vol. 16, no. 4, pp. 2231–2247, Apr. 2025.

[36] H. Sun, Z. Yao, B. Du, J. Wan, D. Ren, and L. Tong, "Spatial–frequency residual-guided dynamic perceptual network for remote sensing image haze removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612416, doi: 10.1109/TGRS.2025.3543728.

[37] X. Chen, Z. Liu, T. Huo, S. Duan, and L. Wang, "DDSNet: A dual-domain supervised network for remote sensing image dehazing," *Phys. Scripta*, vol. 100, no. 2, Feb. 2025, Art. no. 025019.

[38] P. Zheng, Y. Zhou, W. Wang, T. Wang, and X. Li, "SAR target recognition through adaptive kernel sparse representation model based on local contrast perception," *Signal Process.*, vol. 223, Oct. 2024, Art. no. 109558, doi: 10.1016/j.sigpro.2024.109558.

[39] T. Sui et al., "U-shaped dual attention vision mamba network for satellite remote sensing single-image dehazing," *Remote Sens.*, vol. 17, no. 6, p. 1055, 2025, doi: 10.3390/rs17061055.

[40] H. Zhou, Y. Wang, W. Peng, X. Guan, and T. Tao, "ScaleViM-PDD: Multi-scale EfficientViM with physical decoupling and dual-domain fusion for remote sensing image dehazing," *Remote Sens.*, vol. 17, no. 15, p. 2664, 2025.

[41] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685, doi: 10.1109/CVPR52688.2022.01039.

[42] J. Xiong, X. Yan, Y. Wang, W. Zhao, X.-P. Zhang, and M. Wei, "RSHazeDiff: A unified Fourier-aware diffusion model for remote sensing image dehazing," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 1, pp. 1055–1070, Jan. 2025, doi: 10.1109/TITS.2024.3487972.

[43] Y. Huang, Z. Lin, S. Xiong, and T. Sun, "Diffusion models based null-space learning for remote sensing image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3370595.

[44] J. Liu, Q. Wang, H. Fan, Y. Wang, Y. Tang, and L. Qu, "Residual denoising diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2773–2783.

[45] Z. Luo, F. K. Gustafsson, Z. Zhao, J. Sjölund, and T. B. Schön, "Refusion: Enabling large-size realistic image restoration with latent-space diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1680–1691.

[46] Y. Wen, T. Gao, J. Zhang, Z. Li, and T. Chen, "Encoder-free multiaxis physics-aware fusion network for remote sensing imagedehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4705915, doi: 10.1109/TGRS.2023.3325927.

[47] Y. Huang, Z. Li, C. Yang, F. Sun, and Y. Song, "SateHaze1K: A satellite dehazing dataset with haze-level annotations," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jun. 2024, pp. 1–10.

[48] Z.-H. Zhu, W. Lu, S.-B. Chen, and C. H. Q. Ding, J. Tang, and B. Luo, "Real-world remote sensing image dehazing: Benchmark and baseline," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 4705014, doi: 10.1109/TGRS.2025.3584234.

[49] A. Ali, M. Khan, and L. Zhang, "DHID: A real-world dehazing dataset for image restoration," Tech. Rep., 2024.

[50] X. Zhang, Y. Ren, and W. Yang, "A comprehensive real-world image dehazing benchmark," Tech. Rep., 2021.

[51] Y. Guo et al., "SCANet: Self-paced semi-curricular attention network for non-homogeneous image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1885–1894.

[52] S. Nedungatt, K. S. Basavaraju, and S. Lal, "MDEANet: Modified detail-enhanced convolution and attention-based network for dehazing of remote sensing images," *Multimedia Tools Appl.*, vol. 84, no. 18, pp. 18943–18966, Jul. 2024, doi: 10.1007/s11042-024-19792-7.

[53] N. Kim, I.-S. Choi, S.-S. Han, and C.-S. Jeong, "DA-Net: Dual attention network for haze removal in remote sensing image," *IEEE Access*, vol. 12, pp. 136297–136312, 2024, doi: 10.1109/ACCESS.2024.3459588.

[54] X. Wang, B. Yuan, H. Dong, Q. Hao, and Z. Li, "End-to-end multi-scale adaptive remote sensing image dehazing network," *Sensors*, vol. 25, no. 1, p. 218, Jan. 2025, doi: 10.3390/s25010218.

[55] H. Jin, Z. Chen, Z. Song, and K. Sun, "DFFNet: A dual-domain feature fusion network for single remote sensing image dehazing," *Sensors*, vol. 25, no. 16, p. 5125, Aug. 2025, doi: 10.3390/s25165125.

[56] S. Ahmad, X. Zhang, J. Liu, and Y. Wang, "CLEAR-Net: Cascaded local and external attention for remote-sensing dehazing," *Remote Sens.*, vol. 17, Jan. 2025, Art. no. 2664.

[57] X. Zhang, F. Xie, H. Ding, S. Yan, and Z. Shi, "Proxy and cross-stripes integration transformer for remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3457868.

[58] J. Weickert, *Anisotropic Diffusion in Image Processing*. Stuttgart,Germany: B. G. Teubner, 1998.

[59] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 7, pp. 629–639, Jul. 1990, doi: 10.1109/34.56205.

[60] L. Zhang and S. Wang, "Dense haze removal based on dynamic collaborative inference learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3207832.
[61] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Computer Vision—ECCV 2016*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds., Cham, Switzerland: Springer, 2016, pp. 694–711, doi: 10.1007/978-3-319-46475-6_43.
[62] X. Cong, J. Gui, J. Zhang, J. Hou, and H. Shen, "A semi-supervised nighttime dehazing baseline with spatial-frequency aware and realistic brightness constraint," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2631–2640.
[63] N. Liu et al., "Winscaleformer: Diffusion-attention-based single remote sensing image dehazing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 28498–28514, 2025, doi: 10.1109/JSTARS.2025.3623240.
[64] J. Li, K. Chi, Y. Chang, and Q. Wang, "DFG-DDM: Deep frequency-guided denoising diffusion model for remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, doi: 10.1109/TGRS.2025.3585894.
[65] H. Zhou, Y. Wang, Q. Zhang, T. Tao, and W. Ren, "A dual-stage residual diffusion model with perceptual decoding for remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4109312, doi: 10.1109/TGRS.2025.3600540.
[66] J. Ning, J. Yin, F. Deng, and L. Xie, "MABDT: Multi-scale attention boosted deformable transformer for remote sensing image dehazing," *Signal Process.*, vol. 229, Apr. 2025, Art. no. 109768.
[67] Z. Lihe, J. He, Q. Yuan, X. Jin, Y. Xiao, and L. Zhang, "PhDnet: A novel physic-aware dehazing network for remote sensing images," *Inf. Fusion*, vol. 106, Jun. 2024, Art. no. 102277.
[68] T. Cui et al., "Physics-driven image dehazing from the perspective of unmanned aerial vehicles," *Electronics*, vol. 13, no. 21, p. 4186, 2024, doi: 10.3390/electronics13214186.
[69] Z. Yu, C. Liu, L. Liu, Z. Shi, and Z. Zou, "MetaEarth: A generative foundation model for global-scale remote sensing image generation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1764–1781, Mar. 2025, doi: 10.1109/TPAMI.2024.3507010.
[70] Y. Li et al., "Full-scope vectorization of geographical elements from large-size remote sensing imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Feb. 3, 2026, doi: 10.1109/TPAMI.2026.3660934.



Naveed Sultan received the B.S. degree in information technology from the Khwaja Fareed University of Engineering and Information Technology, Rahim Yar Khan, Pakistan, in 2021, and the M.Eng. degree in electrical engineering specializing in data analytics from Chulalongkorn University, Bangkok, Thailand, in 2024.

He is currently a Researcher with the Department of Computer Engineering, King Mongkut's University of Technology Thonburi, Bangkok. His work centers on advancing image super-resolution using deep learning and transformer architectures. His research has been presented at prominent international venues and continues to drive progress in real-world applications of super-resolution modeling. His research interests include super-resolution, change detection, object detection, and semantic segmentation.



Mansoor Hayat (Member, IEEE) received the B.Sc. degree in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2015, the M.Sc. degree in electrical engineering from the Institute of Southern Punjab, Multan, Pakistan, in 2018, the M.B.A. degree from the National College of Business Administration and Economics, Lahore, Pakistan, in 2021, and the Ph.D. degree in electrical engineering from Chulalongkorn University, Bangkok, Thailand, in 2024.

He is currently a Post-Doctoral Fellow with the Section of Neurosurgery, Department of Surgery, Rady Faculty of Health Sciences, University of Manitoba, Winnipeg, Canada. His research interests include deep learning and machine learning for medical imaging and video analysis, with a focus on segmentation, reconstruction, and interpretable models.

Dr. Hayat received the Best Conference Paper Award from the 2023 IEEE Region 10 Conference (TENCON), Chiang Mai, Thailand.



Santitham Prom-on (Member, IEEE) received the B.Eng. degree in computer engineering and the Ph.D. degree in electrical and computer engineering from the King Mongkut's University of Technology Thonburi, Bangkok, Thailand, in 2002 and 2009, respectively.

He is currently an Assistant Professor with the Department of Computer Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi. He has authored or co-authored over 50 publications, including high-profile journal articles and conference papers in areas such as speech modeling, deep articulatory synthesis, signal processing, and Thai language processing. His research interests include big data analytics, spatial data analysis, signal processing, and computational modeling.